

Universe24: how a simple rule and the group of 48 arrive at Einstein and Lorentz

Alon Gonen

Independent researcher, Haifa, Israel

alon@defounder.ai ORCID 0009-0008-2701-4599

October 2026

Abstract

We take the world to be the meeting of the past and the future: that meeting is a click, and an observer sees only clicks. One local rule in whole numbers, Rule3, steps every Node of a cubic lattice each interval from its present, its past and its six neighbours, by one Euclidean division whose remainder stays at the Node, so every run is exactly reversible. The board holds only events, split to the six neighbours at every Node: forward from an emitter they are the future, every path at once; read backward from a realised click by the same rule, the past. A click is their meeting at a declared detector, a region of Nodes, and which meeting is realised is the one non-local element. A family enters as one row, its pair and what it holds, and the universe has as few free numbers as the law allows. Gravity is a level every record writes and reads, slowing a Node's clock once and its Links twice; from the coefficients come the gravitational redshift, light's bending by twice the clock's share, the perihelion's advance of 6π times the potential, Lorentz's factor at a body's own light speed, and the force between parallel moving charges as Coulomb's over that factor. Running no experiment, the paper derives what two must give: the two-slit fringe row, and Bell's correlation $E = \cos 2(a - b)$ with the Clauser-Horne-Shimony-Holt sum $2\sqrt{2}$, where every local credit gives at most 2.

Keywords: integer lattice rule; reversible cellular automaton; measurement as a meeting of past and future; Bell's inequality; weak-field gravity; discrete relativity

1 Introduction

The question. The law of this paper is one line of integer arithmetic applied at every Node of a cubic lattice at every interval. Nothing else acts: no field equation beside it, no law of motion for a body, no rate, no draw, no real number. The paper states that line, proves what is exact on the lattice, derives what follows from the line alone, names what is decided beside it, and sets the law's own numbers beside the forms of nature: the forms of Einstein, Lorentz, Newton, de Broglie and Young appear only on the comparison side, beside the law's expectation written before any run; none is an input. The group of the title is the 48 signed permutations of the three axes, the cube's full symmetry group, its 24 rotations with their reflections, under which the law is covariant with the translations; Einstein and Lorentz in the title are their forms, the weak field's bending, delay, redshift and perihelion and Lorentz's factor for a moving clock, arrived at from the rule's coefficients and never its equations or its group as inputs. Every formula is taken from the law's document [1] and cited by the name of the section it stands in; the engine that computes the rule is described by its own document [2], and the decisions in force by theirs [3]. The simulator, the

engine Universe24, is the law's implementation and nothing else: every section below says in one sentence where its line lives in the engine, and Section 9 says in a few lines how to run it, so that the algebra of the paper and the simulator are one thing from the first line to the last.

What our world is. The owner's statement, which the paper carries as its first sentence and never contradicts (ASSUMPTION, the owner's word of 2026-10-01): our world is the meeting between the past and the future; that meeting is a click; humans see only clicks; the law below is an implementation of that meeting, and from it come many things that nature shows, which physics states without deriving them from anything deeper, and which the one rule gives. In the law's own numbers the statement reads so. A Node holds a record's present and its past, the two levels now and before; Rule3 makes its future, next, from the two and the six neighbours, and nothing else happens on the board. The form read at the Node, $D = \text{now}^2 - \text{next} \times \text{before}$, is the present against the product of the future and the past; it sources every field and it is the share a detector reads. The board itself holds nothing but events that split and their sums: at every Node, every interval, a level splits by Rule3 to the six neighbours, and the record is the sum of the split events. The record runs in both directions of time by the one line, and in both it is these split events: forward from the emitter, every path at once, the future; backward from a realised click, Rule3 at -1 , the same splitting read back to the emitter, what happened, the past; through two slits both pass both gaps, a beam that splits into two fans (Section 3.2). A click is where the two carry the same signal at a declared detector; the detector asks only whether the future and the past met at it, and which meeting is realised is the draw, the one non-local place of the description and the one thing the law does not know (Section 3.2). A Node in the now reads neither the next interval nor the realised path, so no Node is read from inside the world; a Node is read only through a detector, and every number of the board we print is ours from outside the world, a diagnostic that nothing in the world has (Section 1). The reversibility of the board, exact to the bit, is the law's own reason and no convenience of the engine: a past read back from a click needs it, and a pair's root is reached through it. What follows from the one rule, each in its section and none an input: the bending of light by twice the clock's share and the perihelion's 6π (Section 6); Lorentz's factor at a body's own light speed, with no boost among the lattice's symmetries; the magnetic force as Coulomb over that factor, from the clock once and the Link twice (Section 6, (b2)); the equivalence principle for the bodies of nature (Section 5.3); the two slits' fringe row and the lattice's own arrival interval from the rule's line in real arithmetic (Section 7); the measurement without a collapse, the record never ending and the click a reading (Section 3.8); the uncertainty principle as the rule that no click names a Node (Section 3.3); the arrow of time as the sequence of clicks on a reversible board; a cosmological constant fixed at 0 by one click of nature (Section 4.6); and Bell's correlation [4] through a pair's root with no signal between the sides, derived under the one form of the click (Section 7.3).

The click's story, the spine of the paper. On the board there are events and amplitudes: an emitter's events, declared in a file or written by a body, split by Rule3 to the six neighbours at every Node, every possibility at once, and the amplitudes are their sums at the Nodes, the record. Nothing on the board is a wave and nothing on the board is a quantum: the record is deterministic, bit-reversible, and never ends. A click is a detector's. A detector is a declared region of Nodes, never one Node, at least half a wavelength across, which sees the record's current through its boundary; the quantum's action measures that inflow, so the detector saw N quanta where its inflow over the passage is N walls; and the instrument credits each quantum to one detector, by the shares of the inflows over every declared detector, one click per quantum and never two (Section 3.3). The click is the meeting of the future with the past at that region, and the credit

is the draw among the meetings: a declaration of the instrument and no line of the law, the one non-local place (Section 3.2). The record goes on after the click; what a detector absorbed leaves the declared board through the face that backs it; the other possibilities go on as empty amplitudes, in no click (Section 3.8). Two lines that moved whole quanta and the sign between Nodes by a division at every Node, the count's line and the sense's line, left the law at the owner's word of 2026-09-30: the count is the record's share, a reading, and nothing stands at a Node but the record and the one write's remainder (Section 3.4). The price of locality is stated plainly in Section 3.9: exclusivity and Born's rule [5] are the instrument's and no finding; a detector in one gap of the two slits that absorbs is the closed slit; a local credit of a pair's classical phase gives Bell's bound 2 exactly [6], and the meeting through the pair's root is what reaches quantum theory's $2\sqrt{2} = 2.83$ with every quantum counted, derived and not run. A finding against nature is a click compared with a click, nothing else: where a click row of the law stands beside a click row of nature's and differs, the paper writes it as it stands; a number computed from the algebra and set beside nature's number is a computation, marked so; and a GameBoard reading is never compared with nature.

The reading rule. Inside is inside the GameBoard, where no one measures; Outside is the game above the board, the detectors and their clicks. Only a detector's click is a measurement, compared with nature or pinned as an expectation. A number of the GameBoard's state (a level, a support, a total, a body's centre, a region's inflow before the credit) is a GameBoard reading, a diagnostic, labelled so wherever it appears, never compared with nature and never a result on its own; a number whose kind is not named is not a result. The reason is the law's own: a Node in the now holds one direction only, now and before giving next, and sees neither the future (the possibilities not yet realised) nor the past (the realised path, which exists only as the backward reading from a click); a reading is the coincidence of the two directions, and they coincide only at a click; so a Node of the board is read only through a detector, and a diagnostic is ours from outside the world, which nothing in the world has [3, a Node of the board is read only through a detector]. So clicks are compared with clicks: nobody measures the board, and a finding against nature is a click row of the law beside a click row of nature's, never a reading of the board and never a computation on its own [1, readings and measurements; the reading][3, only clicks against clicks]. The paper names no wave on the board: where nature says a wave the paper says an emitter's events that spread and a detector that measures them or not, and between the two ends events and amplitudes; where nature says a plane wave it says a plane record, the record of one wave number; interference is split events meeting with opposite signs (Section 3.5). What produces a click is a meeting, and which meeting is realised is not known and not claimed: everything on the board is deterministic, a human measures only realised events, and the detector's declaration stands in the place of a mechanism for the choice, as nature's detectors are described by what they report and not by why [1, what produces a click is not known][3, the click is the meeting of the future with the past]. The paper claims that the Outside represents reality and shows the match; it never claims that nature is so [3, inside and outside].

The marks. Every claim carries one of five marks: THEOREM, proved for every GameBoard; DERIVED, from theorems by argument; COMPUTED, a number of the law's own computed from its lines before any run; ASSUMPTION, a line the law holds by decision and names where it is used; HYPOTHESIS, a line outside the law under its own name, with what would decide it. No number of a run stands in this paper (Section 7); when runs enter, they enter only through the documents' rows, labelled DETECTOR (a click, the measurement) or GAMEBOARD (a diagnostic). The law's document, the engine, the derivations with their checks and the text were worked out with a large language model used as a tool under the author's direction, and the author reviewed every statement; its use

is stated under Statements and Declarations at the end of the paper.

The law in one line. At every Node, each interval, for every family [1, the line; the booking; the primitives]:

Rule3:	$w a_{\text{next}} + r' = \sum_{a \in \{x,y,z\}} R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{ den } \Gamma^2, \quad R_a = 2 \text{ num } p_a^2,$ $S = 12 \text{ den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{ num } (p_x^2 + p_y^2 + p_z^2);$
the one write:	a held part at a Node gains $\left(\sum_f w_f q_f(\text{Node}) + r \right) \text{div } E$, the remainder r kept there;
the readings:	$D_i = \text{now}_i^2 - \text{next}_i \text{before}_i, \quad W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}},$ $F_{ij} = \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j).$

(1)

In the first line a_{now} , a_{before} and a_{next} are the family's level at the Node now, one interval before and one interval on, the present, the past and the future; $\text{arr}_{\pm a}$ is the arrival through the Port $\pm a$, the neighbour's level; r and r' are the remainder kept at the Node before and after the step; Γ is the Node clock, the pace of an empty Node; $[\text{num}, \text{den}]$ is the family's pair, with $\cos \omega_0 = \text{num}/\text{den}$ its rotation per interval at wave number zero; p_0 is the Node's clock pace and p_a its pace along the axis a , both the Node clock lowered by the levels the family reads there; and div is the floor division. The second line is every write into a held part of a held family's level, one division per part per interval: the sum over the sourcing families f of the writer's own quantity there, q_f , each at the one weight w_f with which that family reads the written family, plus the part's one remainder, over the part's wall E , $E_s T$ for the time part and $E_s W_c$ for an axis part, E_s the row's divisor. The third line is what is read from a record and never written: its form D_i at a Node, the present against the product of the future and the past, whose quotient over the quantum's action T is the well that sources the fields; the Wronskian W_i of its two level pairs, the sign; and its current F_{ij} into the Node i from its neighbour j over the interval, Rule3's own conserved flux, which a declared detector reads through its boundary and which the instrument counts in quanta, one per wall $W_c = 3 \text{ den } T$ of inflow. Everything else is a number in a file. The first line is a theorem's object (Section 2), the second the one write (Section 2.8), the third the click's (Section 3).

The words. The physical location is a *Node*; its complete local information is its *NodeState*, the law's own numbers and nothing else. Nodes are joined by *Links* through six directional *Ports* per Node; the lattice of Nodes is the *GameBoard* (Fig. 1); a local change is an *Event*, and there is one *LocalRule*, Rule3. A *Family* is one kind of physical value with its pair and what it holds; its *Record* is its levels over the GameBoard, two per Node with a remainder. A *Body* is the Nodes on which a family's record stands bound, its quanta the record's share, with no law of its own; a *Message* is a record the world lays and no body, an emitter's events. A *Detector* is a region of Nodes declared in the file to report, a declared instrument; its *Click*, one quantum credited to it by the instrument from the current it saw through its boundary, is the one measurement, reported with the region, the family and the Ports crossed and never a Node. An *Interval* is one step of the whole GameBoard [1, the objects][2, the words].

What this paper claims, and what it does not. Claimed: the owner's statement of what the world is, carried as the paper's first sentence; one explicitly stated integer rule whose reversibility, conserved form and momentum's flux are theorems on the lattice; the click as the meeting of the future with the past at a declared detector, the record never ending, with the one non-local place named; the families as bands of that rule, at a Node that knows no family and no direction of time; the cloud as the body of nature, and the compact pixel as collapsed matter; the forms of nature

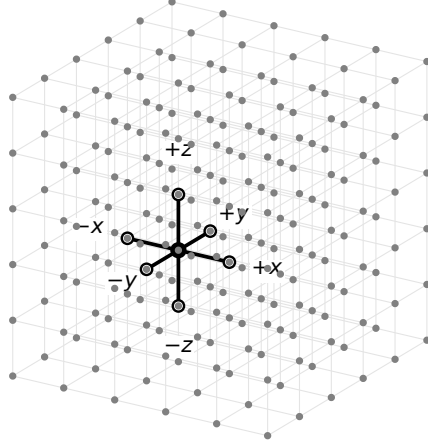


Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x$, $\pm y$, $\pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Fig. 2. A Node keeps each family's two levels and its remainders there, and nothing else. Drawn from the definitions; no run.

arrived at from the coefficients and set beside their known forms; and two experiments named and derived, the two slits and Bell, each as the law's own statement of what it must give, no run's number among them. Not claimed: that nature is so; a derivation of the Lorentz group, which is no symmetry of the lattice (the lattice's symmetries are the 48 signed permutations of the axes); which meeting becomes the click; any run's number, until a run has worked and been confirmed; the fine-structure constant, the mass ratio of the proton to the electron or what fixes the Node clock, which are open by name (Section 8.4). The paper holds no run of its own; every number of a run is the documents' and is cited so.

2 The general formula

This section states the object and proves what is exact on it: the lattice and its families, the one line with its coefficients, its reversibility, what its acts can and cannot form, the paces with their guard, the conserved form, the tension as the wave's own stress, the one write, the interval and its inverse, and the three channels through which one family acts on another [1, the GameBoard and the families; Rule3; the primitives].

2.1 The objects

A Node is a physical location. The GameBoard is all the Nodes together, a box of (N_x, N_y, N_z) Nodes; each axis is periodic (the box a torus on that axis), open (a face beyond which no Node lies, read as 0) or folded (an axis of one layer, whose two Ports return the Node itself), as the world declares. Two Nodes that differ by one step along one axis are joined by a Link; each Node has six Links, one through each of its six Ports, $\pm x$, $\pm y$, $\pm z$ (Fig. 2). An inner face is the face's rule declared inside the board: a plane of Nodes across an axis, beyond the board but for the gaps the world declares; a Node beyond the board is read as 0 through every Port and reads 0 through its own (a held row with a declared rest is read at its rest there, 0 since the vacuum content is 0, the

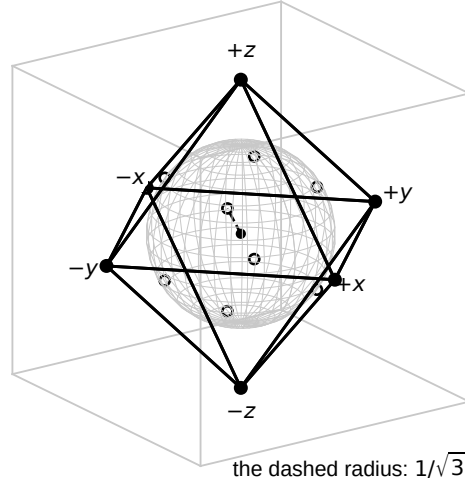


Fig. 2 The six Ports of one Node, $\pm x$, $\pm y$, $\pm z$, and the octahedron they span: the Nodes one interval away. Its inscribed sphere, of radius $1/\sqrt{3}$ (the dashed radius), touches the eight faces on the cube's diagonals (the open circles), and that radius is light's speed at long wavelength, $1/\sqrt{3}$ Links per interval [1, the constants]. The maps of the six Ports onto themselves that keep opposite Ports opposite are the 48 signed permutations of the axes, the symmetry group of the cube drawn faint behind it, under which the law is covariant. Drawn from the definitions; no run.

vacuum beyond the face being the same vacuum), so no level, current or source crosses its Links, and no body, message or detector stands there. The board's shape is a declaration of the file, like the wraps and the detectors' regions, and no number of the engine. A face may also be declared receding (ASSUMPTION, the owner's decision, built): the board is a dense cube that grows by a layer of zeros on that side whenever a level other than 0 reaches the Node before the face, so no event ever meets it, and a run of such a board is a run of the unbounded lattice with 0 beyond the front, since Rule3 needs nothing else and a level leaves 0 only where a neighbour was not 0 the interval before; an option beside open, folded and periodic faces, never the default, with a declared largest size at which a run reports and ends; on it the massless and the binding rows build their rest behind the front by Rule3 itself, Huygens' construction on the lattice, nothing reflects, the blind numbers of the looks are the infinite lattice's exactly, and the world's size is its age in Links; a receding face grows the board and stretches nothing, so Hubble's redshift does not come from the law's growth [1, the unbounded board][3, the receding face]. The local information of a Node is its NodeState: per family, per part, two levels and Rule3's remainder, and per held part the one remainder of its write, and nothing else; its count, its sense, its currents and its tension are readings of the record; no register, no table, no formula and no draw stand at a Node [1, the objects][3, nothing at a Node but the law's numbers].

A family is one kind of physical value with its own declarations: its pair [num, den] and what it holds. Its record is two levels per Node (now and before) with a remainder per Node; a family of quanta carries two such level pairs, the real one and its second, the rotation sense (Section 3.6). A body is a set of Nodes on which a family's record stands bound, one connected region, its quanta the record's share there (Section 5); a message is a record the world lays and no body, a packet of a family of quanta with its axis, its wave number, its amplitude and its envelope declared [1, the objects].

2.2 The line

At every Node, each interval, every family steps by the first line of Eq. (1): the six arrivals, each the neighbour's level through its Port (0 beyond an open face, the Node's own level on a folded axis), the Node's own level now, minus its level before, plus the remainder kept at the Node, divided once by the wall w ; the quotient is the next level and the remainder stays. The remainder is the Node's own and never travels [1, the line].

The name, and the one division. Rule3 is named for the three axes of the lattice, the three directions of the vector the rule works with: the Link's paces (p_x, p_y, p_z) are that vector, and the line has one read coefficient per direction, $R_a = 2 \text{ num } p_a^2$, serving the direction's two Ports; the six Ports and the wall's 6 are the three directions twice, so the six follows from the three; in time the rule is a three-level recurrence (before, now, next), and in light's own line the three stands in the open: the next level is the six arrivals over 3 minus the level before (the owner's word of 2026-10-01: the name stays). At every Node, each interval, Rule3 is one division of whole numbers. The six arrivals through the Ports, each weighted by its direction's coefficient, together with the Node's own level, the level before and the remainder the Node kept, are summed as one whole number; that whole is divided by the wall w ; the quotient is the next level, and the remainder, $0 \leq r' < w$, stays at the Node and never travels. The whole is taken apart over the Ports and put together again at the neighbours: nothing is rounded away and nothing is lost, the step back is the same division, and that is why the board is exact and reversible to the bit (the owner's word of 2026-10-01, "you take the whole and take it apart over the Ports and leave the remainders") [1, the line][3, the paper].

In the engine the line is one function, `core/rule3.py`, called on whole-board arrays for every record with the direction $+1$ or -1 , and every level that crosses a Link crosses it through a Port, `core/ports.py`; a test holds the rule's arithmetic to that one function and every integer literal of the engine to the law's own [2, the gates]. At the vacuum's pace $p_0 = p_a = \Gamma$ the self coefficient vanishes and the line is the plain second-order wave rule of the pair,

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

whose rotation at the wave vector $\mathbf{k}$ is $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$: at wave number zero $\cos \omega_0 = \text{num}/\text{den}$, and for the massless pair [1, 1] along one axis $\cos \omega = (2 + \cos k)/3$, so that $\omega = |k|/\sqrt{3}$ as $k \rightarrow 0$. Light's speed at long wavelength is therefore $1/\sqrt{3}$ Links per interval, under the causal bound of one Link per interval, the arrival's own speed; the two differ, and what in nature is one Link and one interval rests on the units of Section 8 [1, the lattice constants].

The clock once and the Link twice. From the coefficients, the rotation at a Node of the clock's pace p_0 and the Link's paces p_a is (DERIVED)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{ den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a), \quad (3)$$

the mass term scaling with the clock's square and the band term with the Link's pace. Two readings follow and no third [1, the clock once and the Link twice]. (i) *The clocks shift alike*: at $k = 0$ every massive band's rest rotation obeys $1 - \cos \omega_b = (p_0/\Gamma)^2 (1 - \cos \omega_0)$, one factor for every family, so two bound bodies at one Node shift their clocks alike, and a packet's fall in a pace gradient does not depend on its family beyond its band's own shape. (ii) *The Link is twice*: along a Link every wave's speed falls with $p_a = \Gamma - 2c - t_a$, the held level c entering the Link twice and the clock once, so light, which reads the held rows as every family does, is delayed and bent by twice the clock's share

(Section 6). The line is not conformal: a light wave at a fixed wave number scales with $(p_a/\Gamma)^2$ and a bound mode at rest with $(p_0/\Gamma)^2$, and the two differ already at first order in c/Γ .

A held family's field steps at its row's pair with the pace 1 and the wall 3 den, at first order: for the massless row $a_{\text{next}} = (S_6(a_{\text{now}}) - 3 a_{\text{before}} + r) \text{ div } 3$, S_6 the sum over the six neighbours, and for a row with a gap 3 den $a_{\text{next}} + r' = \text{num } S_6(a_{\text{now}}) - 3 \text{ den } a_{\text{before}} + r$, the same call at the row's own pair [1, the vacuum's row steps at the divisor 1].

2.3 The direction

Let σ be the direction, +1 forward and -1 backward, let div and mod be the floor division and its remainder in $[0, w)$, and let ρ be the remainder carried in. With

$$u = \sigma \left(\sum_a R_a \text{arr}_a + S x - w y \right) + \rho, \quad z = \sigma (u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (4)$$

forward $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$ gives $(z, \rho') = (a_{\text{next}}, r')$, and backward $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$ gives $(z, \rho') = (a_{\text{before}}, r)$.

Theorem 1 (the direction). *The backward call returns (a_{before}, r) bit for bit; the reversibility of every step needs no second function.*

Proof. Forward, $\sigma = 1$, the line is the definition of div and mod . Backward, $\sigma = -1$: write Y for the right side's first two terms, $\sum_a R_a \text{arr}_a + S a_{\text{now}}$. Then $u = w a_{\text{next}} - Y + r'$ and $z = -\lfloor u/w \rfloor = \lceil (Y - w a_{\text{next}} - r')/w \rceil$. From the forward line, $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$ with $0 \leq r < w$, whose ceiling divided by w is a_{before} exactly; and $\rho' = u - w \lfloor u/w \rfloor = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$. Checked bit for bit on 20,000 random cases against a forward and a backward function written separately [1, the direction]. $\square$

2.4 The four acts and what they can form

Rule3's line is one act; declared coefficients make it four: (a) the read, the line over the GameBoard with its six arrivals; (b) the one-Node step, the line with the neighbours' reads declared 0, $a_{\text{next}} + a_{\text{before}} = (a/b) a_{\text{now}}$, the rotation by the angle whose doubled cosine is a/b (Chebyshev's recurrence), which with $a/b = 1$ and the coefficient on a_{before} declared 0 keeps a level and with $a/b = 2$ from $(0, 1)$ counts; (c) the division, the line with $R = S = 0$, Euclid's division with the remainder kept; (d) the load, a declared integer in the numerator, how a count enters a level. A composition of the acts is a finite sequence of them, each with its coefficients declared in the files, applied to declared levels [1, the four acts].

Theorem 2 (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

Proof. Each act is a linear form of the levels or the floor of one; a floor of a linear form is piecewise linear with integer slopes; sums and compositions of such functions are of the same kind. The second difference of such a function in any one level is 0 almost everywhere, while the second difference of a product $\text{now}_i \text{before}_j$ in the pair $(\text{now}_i, \text{before}_j)$ is 1. So nothing bilinear comes from the acts alone. $\square$

What the theorem separates: every product of two levels in the engine is a *booking*, a reading of a bilinear form with declared coefficients and no write (Section 3.1); the three bookings the law

reads are the form, the Wronskian and the current, and the current is Rule3's own conserved flux (Section 2.6). Nothing on the board is written from a product; a detector's reading of the current is the host's and writes nothing.

The three tests and the six verbs. A line enters the law only if it is generic (one primitive with declared integers, no family's name and no kind), vector (one of the acts on the state, no root and no float) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers); a hypothesis that needs more is stated under its own name, outside the law. Every integer of the law is made of six operations only: the translation of an accumulator by its rate, the bilinear form with a declared matrix, the group-ring addition, the permutation, the evaluation at the primitive root of unity, and the Euclidean division with the remainder kept; a root, a true division or a draw is outside the law. No act of the interval takes a square root: the integer square root stays only at load and in the generator, as the fixed point of the division act iterated, $x \leftarrow (x + n \text{ div } x) \text{ div } 2$ from above until it repeats [1, the three tests of every rule][3, the six verbs; the root leaves everywhere].

2.5 The paces and the guard

The pace of a Node for a family is what its reads make of the Node clock:

$$p_0^2 = (\Gamma - c)^2 + c^2, \quad p_a = \Gamma - 2c - t_a, \quad c = \sum_{\text{reads}} \text{weight} \times \text{by} \times \text{the read family's time level}, \quad (5)$$

t_a the axis content from the read families' tensions on the axis a (each $\text{weight} \times \text{by} \times \text{the } aa \text{ part div } 2$, one division per read per axis, rounded at the read since the pace is a coefficient and no level). A positive weight is a hollow (the read family's level slows the clock, an attraction); a negative weight is a hill. By sign the weight is multiplied by the reader's q at the Node, the sign of its own rotation sense there (Section 3.6). A family with no reads steps at $p_0 = p_a = \Gamma$, the plain rule. The clock's slowing is this pace: a record at a Node of pace p rotates and moves as a record at Γ does, with its intervals p/Γ as long. Every family of quanta reads every holder of the content at the weight 1, a row without a gap and a row with a gap alike as its stepped field, $x = (L_b + L_g)/\Gamma$ with L_b the binding holder's level and L_g gravity's, and every holder of the sign by its own q ; a family sources what it reads at the weight it reads with [1, the paces; the reads are the held row's pair's].

The clock's second order is the composition. Each unit of the level slows the clock as it already stands, $p_0 = \Gamma(1 - 1/\Gamma)^c$, whose square to the second order is $(\Gamma - c)^2 + c^2 = \Gamma^2(1 - 2\Phi + 2\Phi^2)$ in the level's share $\Phi = c/\Gamma$; Rule3's coefficients take the paces' squares as written, with no division act, no rounding and no root, and $p_0 = \Gamma - c + c^2 \text{ div } 2\Gamma$ to the unit is a reading of the algebra; the Link's pace stays $\Gamma - 2c - t_a$, the level entering it twice [1, the clock's second order is the composition]. Section 6 reads the perihelion off this second order.

The guard, on squares, once at load. $0 < p_a$ and $p^2(\text{den} + \text{num}) \leq 2 \text{ den } \Gamma^2$ at every Node, with the edge's square $P^2 = 2 \text{ den } \Gamma^2 \text{ div } (\text{den} + \text{num})$, Γ^2 for light and above it where $\text{num} < \text{den}$, read on the clock's square and on each Link's pace alike over the whole initial state and never in the interval. Below 0 a pace is no clock; beyond P the step is unstable: the mode at wave number π on every axis has the factor $2 - 2(1 - \text{num}/\text{den})(p_0/\Gamma)^2 - 4(\text{num}/\text{den})(p_a/\Gamma)^2$, which is $-2 \text{ num}/\text{den}$ in the vacuum and stays at or above -2 wherever p_0 and p_a are both at or below P , since $(P/\Gamma)^2(2 - 2 \text{ num}/\text{den} + 4 \text{ num}/\text{den}) \leq 4$ exactly when $P^2(\text{den} + \text{num}) \leq 2 \text{ den } \Gamma^2$ (THEOREM). In the run no act reads a Node to refuse: every coefficient is a square, $R_a = 2 \text{ num } p_a^2$ passes through 0 as the content reaches $\Gamma \text{ div } 2$ (the Link closes: the horizon) and reopens mirrored beyond it, and

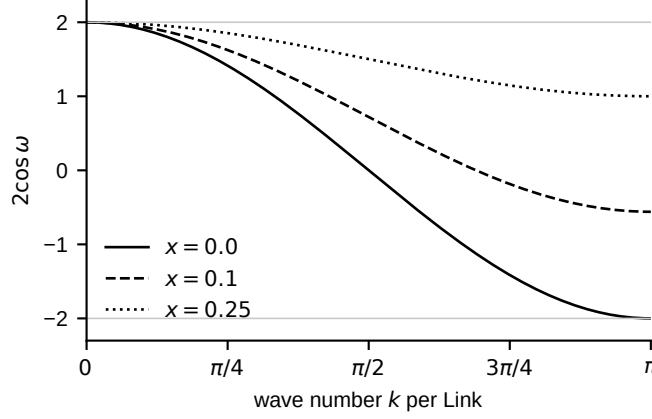


Fig. 3 Light's band at a content $x = L/\Gamma$ read into the paces (DERIVED, Eq. (3) at num = den): the doubled cosine of the rotation along one axis against the wave number, $2 \cos \omega = 2 - 2(1 - 2x)^2(1 - \cos k)$, at $x = 0, 0.1$ and 0.25 . The band's bottom rises with the content, its top stays at 2; at the horizon $x = 1/2$ the band closes to a point. The slope at $k = 0$ is light's speed, $(1 - 2x)/\sqrt{3}$ Links per interval. Drawn from the definitions; no run.

beyond P the mode at π grows until the host's trap on the arithmetic's width ends the run; the report gives the pace's minimum from the final state as a diagnostic [1, the guard][3, no one asks a Node; the root leaves the run].

The band at a pace. A reading of the line and no new rule: at a Node of paces p_0 and p_a a record's rotation at the wave number k along an axis is Eq. (3), so the band of rotations a body's Nodes carry narrows with the paces; a record whose rotation lies outside the band there is evanescent inside and reflected whole, and one whose rotation lies inside is slowed to the band's group velocity. Light, [1, 1], meets a total mirror where $p_a < \Gamma \sin(k/2)$, and a window otherwise [1, the band at a pace]. At a Node of content $x = L/\Gamma$ light's line has $S/w = 2 - 2(1 - 2x)^2$ and $R_a/w = (1 - 2x)^2/3$, its band is $[2 - 4(1 - 2x)^2, 2]$, and at long wavelength $\omega = (1 - 2x)|k|/\sqrt{3}$: light's speed is $c_0(1 - 2x)$, an index $n = 1/(1 - 2x)$, $n - 1 = 2x$, the content read twice (Fig. 3) [1, the rows against nature, (b)]. At $x = 0$ light's zone-corner mode sits at -2 with no margin, and at any negative content it grows, $\cosh \mu = 2(1 - 2x)^2 - 1$ with μ the growth per interval, so a content below 0 is refused wherever light reads it; the vacuum content, the massless row's declared rest, is 0 (Section 4.6) [1, what is open, item 22].

The lattice constants. The six Ports' own, exact and no number of nature: the massless row's rest around one static source of s quanta per interval at the divisor 1, by the start's iteration on an open box with its faces' constant image removed, is $3G(r)s$ with $3G(0) = 0.7582$ at the Node and 0.2582, 0.1286, 0.0826, 0.0608 and 0.0483 at 1 to 5 Links, toward $3/(4\pi) = 0.2387$ far away (COMPUTED); of which 0.76 s at the Node and 0.24 s/r beyond are the continuum's; and light's $1/\sqrt{3}$ [1, the lattice constants].

2.6 The conserved form

The read matrix $2 \text{ num } p_i^2$ is not symmetric where the paces differ, so the form carries the weight $1/p_i^2$ on the Node terms and the plain current on the Links:

$$E = \sum_i \frac{w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i}{p_i^2} - 2 \text{ num } \sum_i \text{now}_i S_6(\text{before})_i. \quad (6)$$

Theorem 3 (the conserved form). *At constant paces E is exact under the line without the rounding; with the rounding each Node's remainder adds at most one unit of the level per interval to the walk of E . Its Link term, the plain current $\text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ through the Link ij , is the current of Rule3's own conservation law.*

Proof. Without the rounding the line is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ with $\mathbf{M}$ the matrix whose diagonal is S_i/w and whose entries on the six Links are R/w . Let $\mathbf{D}$ be the diagonal matrix of the weights $1/p_i^2$; then $\mathbf{DM}$ is symmetric, since $R_a = 2 \text{ num } p_a^2$ at the reading Node and the weight $1/p_i^2$ cancels it on both sides of every Link. For a symmetric $\mathbf{K} = \mathbf{DM}$ the quantity $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D} \mathbf{x} + \mathbf{y} \cdot \mathbf{D} \mathbf{y} - \mathbf{x} \cdot \mathbf{K} \mathbf{y}$ satisfies $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$: with $\mathbf{a}_{\text{next}} = \mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$, $\mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K} \mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}}$, which is the other side. Multiplied by w this is E up to the Link term's sign convention [1, the conserved form]. $\square$

E is a GameBoard reading. It changes by the work term where the paces move: the total form $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$ of a record under paces that change in time obeys exactly $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}} \rangle$ (THEOREM, by substitution), 0 at fixed paces and negative where a well deepens; the record's share of quanta, read from the same form in the vacuum's coefficients, is kept meanwhile: the mass defect is the form's fall at a kept count (Section 5) [1, the mass defect is the form's fall at a kept count].

2.7 The tension: the momentum's flux is the wave's own stress

The momentum density along the axis a at the Node i is the current per axis, $P_a(i) = (F_{i,i+a} - F_{i,i-a})/\text{num}$, and Rule3's line gives exactly, at constant paces (THEOREM),

$$w [P_a(t+1) - P_a(t)] = \sum_b R_b [G_{ab}(i-b) - G_{ab}(i)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

$G_{ab}(i)$ the stress booking on the b -Link $(i, i+b)$ at one time, the levels now alone; the proof substitutes $w \text{ now}' = \sum_b R_b (\text{now}_{i+b} + \text{now}_{i-b}) + S \text{ now}_i - w \text{ before}_i$ into $w P_a(t+1)$: the S term cancels, the w term is $w P_a(t)$, and the R term is the difference of the fluxes. So the flux of the a -momentum across the b -Link is $-(R_b/w) G_{ab}$, the second booking of the conserved form (its Link term shifted by one Link), and the tension on the axis a at the Node, the part Rule3 reads into the Link's pace and the source of the held row's axis part aa , is the momentum's flux, the mean over the Node's two a -Links,

$$T_{aa}(i) = -\text{num} (G_{aa}(i) + G_{aa}(i-a)) \text{ div } 2, \quad (8)$$

$$G_{aa}(i) = \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}),$$

the neighbour's own difference brought through the Port and no other shift: the tension on a Link is the booking of its two ends' own readings, a level and a difference each, and a Node sources its

tensions with half of each of its Links, so the line is local as F is. With the rounding the identity holds up to the remainders' term, $-r'_i \Delta_{a \text{ now}_i} + \text{now}_i (r'_{i+a} - r'_{i-a})$, exactly. It is read and never written, quadratic in the amplitude as the form is: a plane wave of amplitude b at the wave number k has $G_{ab} = -2b^2 \sin k_a \sin k_b$ exactly, so the flux along the wave is positive, as a wave's pressure is; a uniform record gives 0; a standing body's tail carries a tension that does not vanish in the mean; for a packet of count c and velocity $\mathbf{v}$ it is $c v_a v_b$ within the rounding, the dust's stress with no count in any divisor and nothing undefined at a count 0. With the sign as written a wave's stress deepens the content around it: light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification][3, the tensor part is the wave's own stress].

2.8 The one write

Every write into a held part's level on the GameBoard is one act, the second line of Eq. (1): $(\sum_f w_f q_f(\text{Node}) + r) \text{ div } E$ at a Node each interval, one division per part with the one remainder r carried at that Node, starting at $E \text{ div } 2$; q_f the sourcing family's own quantity there (its form D_f , its Wronskian W_f for a holder of the sign, or its tension $T_{aa,f}$), w_f the one weight of the pair of families (whoever reads with w sources with w), E the part's wall, $E_s T$ for the time part and $E_s W_c$ for an axis part, E_s the row's divisor; the sources are summed in the numerator and never divided apart, so no remainder stands per sourcing family and no well is kept, the well $D_i \text{ div } T$ a reading. The hold and the start are its instances; no number is the act's own. Generic (the rows' keys), vector (one carried division per part), local (the Node) [1, the primitives][3, the one algebra]. The form $D_i = \text{now}_i^2 - \text{next}_i \text{ before}_i$ of a record at a Node is its well's source: for a standing record it is $B^2 \sin^2 \omega_b$ at every interval, B its peak [1, the gate reads the form at any phase].

2.9 The interval and its inverse

The engine steps every family from the state at the interval's start to the state at its end; every read is of the start's values, one Link per interval and nothing in zero time. Before the first interval the whole initial state is checked once against the guard; then the interval has four acts, in order: (i) the read, every family's paces from the held families' levels at the start; (ii) Rule3 on every record, each family of quanta's two level pairs by its pair and its paces, each holder of the content's parts at its own pair at the pace 1; (iii) the readings of every record about the step, its six currents at the pair the step started from, read before Rule3 acts, the tension on each axis from the levels Rule3 left, its form and its Wronskian, and the detectors' reports from the currents, per family and declared region the net inflow through the region's front boundary, the click; (iv) the hold, each held part taking its one write of the summed sources over its wall; light is born here, by the write of a rotating body's varying Wronskian into the sign holder's record [1, the interval][2, the interval].

Backward, the acts run by Rule3's direction -1 : a held row's hold back once every family that sources it is booked back, a family of quanta booked back once its own write is off, so the holder of the sign goes before the rows it sources; every interval returns bit for bit, since every number a family read at the interval's start is one the state after the interval still holds, and the readings follow the record. The back-in-time gate is the whole run: N intervals forward and N back, every array of every family compared bit for bit at every interval on the way back, MATCH or MISS naming the first interval and the first Node that differ; a look is not run before it says MATCH on the look's own world (THEOREM by construction, from Theorem 1, the hold's carried division being its own inverse once every family that sources it is booked back). It runs in the engine's tests on the closed cube, on the chain of a body and on the chain of light alone with its faces receding, and on the

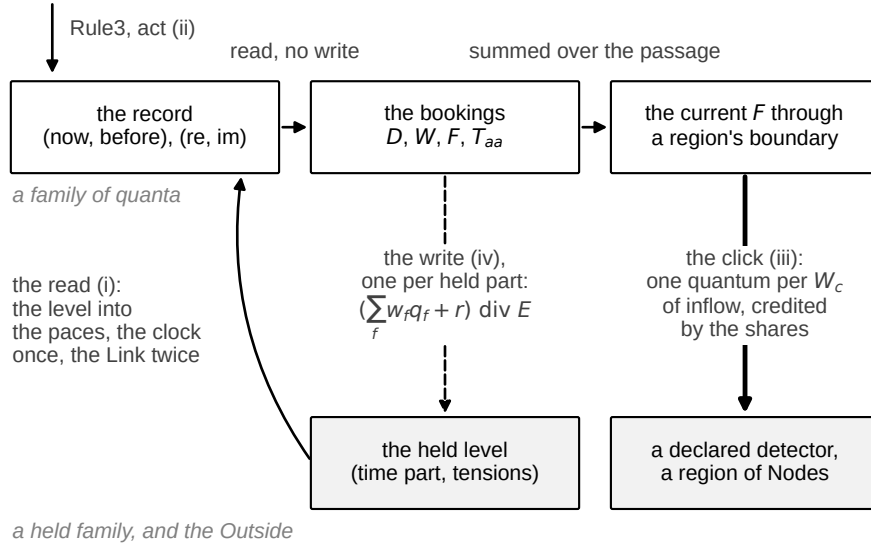


Fig. 4 The three channels through which a family acts on another, drawn from the interval's four acts (Section 2.9). A family of quanta steps its record by Rule3 and books its form and its current; a held family carries a level. The read (i) takes the held level into the record's paces; the write (iv) takes the record's form, its Wronskian's quanta and its tension into the held level over the row's divisor; the click (iii) is a declared region's reading of the current through its boundary, summed over the passage, one quantum per wall W_c of inflow, credited by the instrument; the write (iv) is one division per held part of the summed sources. Every arrow is one line of Eq. (1). Drawn from the definitions; no run.

two experiments' worlds before they run [1, the interval; what is open, item 21][3, the board goes back in time, as a gate]. The exact inverse set one condition on the law's reads: a coefficient read from a Node's own state across a step is not recoverable at the step back. The record's rotation sense, the sign of a two-time Wronskian, is such a coefficient, and where that Wronskian passed zero under a level of the sign holder, at the edge of a charged body, the gate missed; so the law reads a holder of the sign by the family's dimension, a plane reads it and a real line does not, and never by a per-Node sense, and the inverse holds without exception (DERIVED, the owner's word of 2026-10-01 on the advisor's sentence) [3, there is no sense; every family has a dimension].

2.10 The three channels

One family acts on another in three ways and no fourth (Fig. 4): *the read*, at (i), a held row's level times the read's weight into the reader's paces, the clock once and the Link twice, multiplicative, the whole of gravity, of binding and of the charge's pull; *the write*, at (iv), the writer's own quantity through the one weight and the written row's divisor into the held row's level, additive, the source of every field and the birth of light; and *the click*, at (iii), a declared detector's reading of the current through its boundary, the measurement alone, which writes nothing on the board. No product of two levels is written anywhere. No bilinear coupling between families is in the law: the world of clicks (bodies, detectors, the counts) and the world of Nodes (the records) run as one loop and touch only at the click and at a body's Nodes; a record sources the families its own family reads, so every pair of families that read each other is reciprocal, action and reaction one key [3, the coupling is the click alone; light is born by the write].

3 The click

The click is the meeting of the future with the past at a declared detector, and no act of the board: a region of Nodes sees the record's current through its boundary, the quantum's action measures it, and the instrument credits each quantum to one detector. This section states the three bookings the law reads from a record, the meeting as the law, the detector as a declared instrument with its credit, the count as the record's share with the lines that left, the no-wave rule, the sign, the birth of light, what the click ends (nothing), the price of locality and the postulates [1, the click; the reading].

3.1 The three bookings

A booking is a reading of a bilinear form of the levels with declared coefficients; it writes nothing on the board (Theorem 2). The law reads three, the third line of Eq. (1). The form of a record at a Node, $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$, the present against the product of the future and the past, $B^2 \sin^2 \omega_b$ at every interval for a standing record of peak B , whose quotient over T is the record's quanta as the fields' source, the well (Section 2.8). The Wronskian W_i of the record's two level pairs, the sign (Section 3.6). And the current of a record into the Node i from its neighbour j through their Link over an interval, $F_{ij} = \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$, positive inward: a record of wave number k travelling from i to j has $F_{ij} = -2 \text{num} |\psi|^2 \sin \omega \sin k$, below 0. It is antisymmetric, $F_{ji} = -F_{ij}$, and it is the Link term of the conserved form (Theorem 3), the current of Rule3's own conservation law [1, the booking; the conserved form]. A static record has $\text{now} = \text{next} = \text{before}$, so its form is 0 and its current 0 at every Link: a static field carries no quanta and no detector sees it, by the arithmetic and not by a clause. The board holds two quadratic acts and no third: the instrument's reading of the current at a detector's boundary, the click, which writes nothing back; and the hold's write of the form into the held rows over their divisors, the well, a level and no click [3, the generic Node is closed for building].

3.2 The click is the meeting of the future with the past

Assumption 1 (the click is a meeting, the law and not a hypothesis). A record runs on the board in both directions by the one line: forward from the emitter, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at -1 . A click is where the two carry the same signal at a declared detector; its share is the product of the two directions at the region, the form $D = \text{now}^2 - \text{next} \times \text{before}$, the now against the future times the past; which meeting is realised is the draw, the one non-local place. The detector only asks whether the future and the past met at it.

The owner's words of 2026-10-01, verbatim in translation: "the click happens when the past meets the future; the click is the now; the detector only asks whether the future and the past met at it; not a hypothesis; the past and the future propagate on the board, and a click is created only at their meeting" [3, the click is the meeting of the future with the past]. What the backward run is and is not: along the realised path it is the same triples, since the board is reversible, so it brings no new number, carries no count and no signal, and is a reading and not a write; what it does is choose which tree is the past. The meeting through the root: a pair born as one event is two trees with one root; the backward run from the first click passes the root and continues forward on the partner's tree, arriving as the partner's lobe arrives, by linearity a re-weighting of the pair's parts, and the partner's click meets the re-weighted record; each side alone reads one half at every port, nothing is signalled, the detectors read locally and the root joins. The pair family: light's family

with the parts of the pair, two real lines (both lobes through the upper gaps and both through the lower, nature's HH and VV), each part a record stepped by Rule3 alone and never summed at a Node; a source symmetric under rotating both together writes both parts by itself, no labelling declared. One form for the click, and it is the meeting; all the rest are cases of it, one form, one draw, one non-local place (the owner's word of 2026-10-01, 04:20): at the credit the parts are read with the port's coefficients, paired part against the same part through the root with what was realised, summed and squared, the share that density, the draw by the shares; the two slits are the case with nothing to pair (one real line, the share the record's own inflow), Bell the case through the root ($E = \cos 2(a - b)$ and the marginal one half), the absorbing body the case of the resonance when the bodies return, and emission the case with the roles reversed; the two local credits, by the shares and by the signs, are theorems of what the meeting is not [3, the click is the meeting of the future with the past]. In the engine the joint share is computed from the forward records alone: the instrument reads each part's levels at each port at the credit's interval, summed over the port's rows as signed amplitudes, forms $\sum_k c_k A_k(p_A) B_k(p_B)$ over the parts for every pair of ports, takes its density as the joint share and draws once per pair ordered in time, the first side from its marginal and the partner from the conditional; no backward Rule3 is stepped, the root doing its work through the parts' coefficients set at the lay, and the reverse gate stays as the check of reversibility (DERIVED, the advisor's line; the meeting round of the engine, built after this writing, Section 9). Why the inverse in time is required, in the owner's words: because the click is the meeting of the two runs; without the bit-exact inverse the past could not be read back from a click and a pair's root could not be reached, so the reversibility of the board is the law's own reason and no convenience of the engine. A Node of the board is read only through a detector: the now reads neither the future nor the past, the two meet only at a click, and the NodeState's before is the state of a second-order line and no reading of the past (Section 1, the reading rule). What this line replaced: a detector that writes the opposite events behind it (the backward run is a reading, and the absorption stays the receding face); "no joint read" (the detectors read locally and the root joins); and "what produces a click is not known" rephrased, a click is a meeting and which meeting is not known. Unchanged: Rule3's line, the board, no click names a Node, no waves on the event board, the credit for one quantum as the meeting's single-quantum case, and the two slits [3, the click is the meeting of the future with the past; a Node of the board is read only through a detector].

How the meeting looks, schematically. Fig. 5 draws the sentence and no run. (a) *The future covers every possibility.* The emitter's events split by Rule3 to the six neighbours at every Node, every interval; after the wall the record stands at every Node behind both gaps at once, the fans of the figure, and no path is chosen: that is how one rule at one Node covers all the possibilities, by the linearity of the line, the sum of the split events at every Node being the record. (b) *The past is read from the click.* At a declared region of the screen the record's current through the boundary is the share; one wall of the action is one quantum, and the instrument credits a quantum to one region, the click. From that click the same line runs backward, Rule3 at -1 , and retraces the realised path bit for bit, through both gaps to the emitter, the dashed lines: it is a reading of what happened, carries no number the forward run did not and writes nothing; what it does is choose which tree is the past. (c) *Where they meet.* The two directions carry the same signal at the region that clicked and nowhere else; the share there is the product of the two, the form $D = \text{now}^2 - \text{next} \times \text{before}$, the present against the future times the past, and the current F_{ij} is the same product across a Link, the present at one end with the past at the other. In the engine nothing is stepped backward: by linearity the product is computed from the forward records alone, and the gate's MATCH is the proof that the backward reading exists to the bit. (d) *Why two gaps give an alternation and a closed gap a hump.* With both gaps open the split events through the two gaps meet at the screen with the

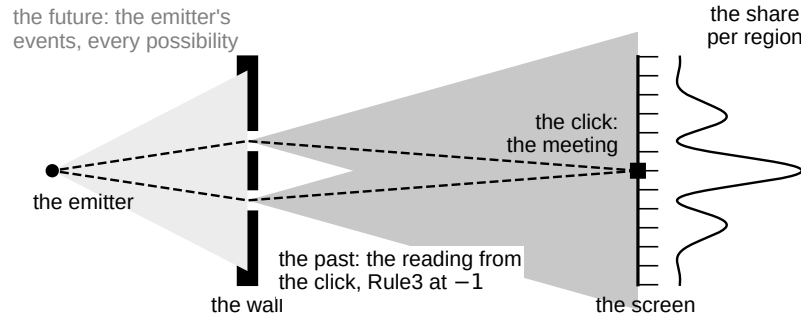
same sign where the path difference is a whole wavelength and with opposite signs where it is half a one, so the share along the screen alternates, the fringes, and the clicks credited by the shares build that histogram quantum by quantum; with an opaque detector in one gap its events leave the declared board through the face that backs it, the backward reading from the screen meets one gap alone, and the share is one gap's envelope with no alternation, a closed slit by construction, as in nature (Section 7.2). (e) *The pair*. A pair born as one event is two trees with one root; the backward reading from the first click passes the root and continues forward on the partner's tree, arriving as the partner's lobe arrives, so the partner's click meets a re-weighted record, by linearity a re-weighting of the pair's parts: the realised click at A's port is the past, a weighting of the parts by A's analyser, read back to the root where the two parts were one event and forward to B where the future is B's analyser's weighting; the meeting at B multiplies the past and the future part by part, the root sums the two branches, and the share is the density of the sum; each side alone reads one half at every port and nothing is signalled. (f) *What a human sees*. Only the click: a Node in the now reads neither direction, the two coincide only at a detector, and every drawing of the board, this one included, is ours from outside the world. Nature's physics knows this picture under two names, the transactional reading of the quantum formalism, an offer wave forward and a confirmation wave backward meeting at the absorber [7], and the two-state vector, a state evolving forward from the preparation and one backward from the outcome [8]; the law's difference is that both directions are one integer line on a board that is exact and bit-reversible, that the product is a reading and not a wave, and that the draw among the meetings is named as the one non-local place and not derived.

3.3 The detector is a declared instrument

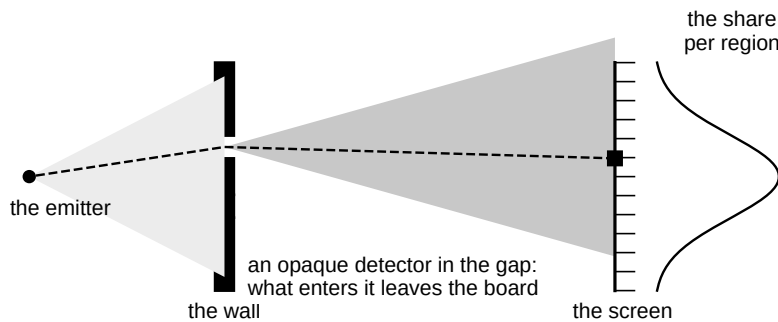
Assumption 2 (the detector's declaration). A detector is a region of Nodes declared in the file, never one Node, at least half the wavelength of the family it reports across the beam (q/p Links for the wave $[p, q]$, 4 at $\lambda = 8$) and one layer deep along it; a narrower region is refused by name at load. It is opaque: backed by a receding face or by the wall's own face, so that what enters it leaves the declared board and never returns, with no act at its Nodes and no log. Its click is its report, per interval and family: the net current into the region through its front boundary Ports, the Ports leading in from the declared board, summed in integers, with the region's name and the family and never a Node; one wall $W_c = 3 \text{ den } T$ of the inflow over a passage is one quantum, so a screen saw N quanta where its regions' inflows sum to $N W_c$. The credit: one quantum to one detector, drawn by the shares of the regions' inflows, the host's reading over every declared detector of the world, one credit per quantum and never two. The credit is the one non-local place of the description and no line of the law.

The owner's words of 2026-09-30 and the advisor's definitions [1, the click is the detector's report; no click names a Node; what produces a click is not known][3, no click names a Node; the two experiments' details under the new laws]. Why a region: a quantum with a direction has no Node, a record confined to one Node being every wave number at once, $\Delta x \Delta k \geq 1/2$ on the lattice as in nature, and the finest structure the amplitudes of a family carry is half its wavelength, two split events of one wave number meeting at the smallest spacing that keeps their signs apart; so no click names a position finer than the region, and every per-Node number of a run is a GameBoard reading. Why the net front inflow: adjacent regions share transverse Links, and a density that entered one region and moved to the next inside the layer is credited where it entered, the click's place; the sum over the regions is the screen's total, and the laid quanta less that total is the part that left elsewhere, reported beside it. Why opaque: a region whose back faces declared Nodes passes the density on and its net inflow reads nothing, and a layer written to 0 reads to its neighbours as

(a) two gaps: the share alternates



(b) one gap closed: one gap's envelope



(c) a pair: the meeting through the root

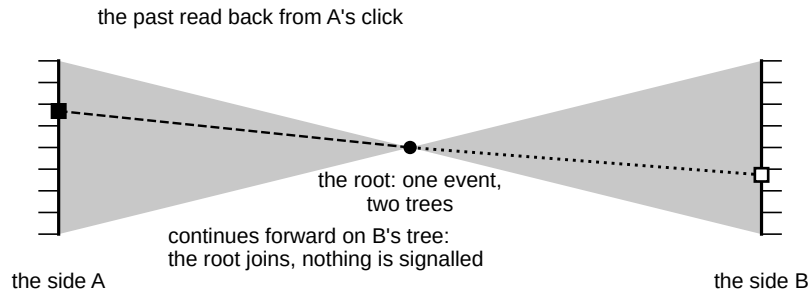


Fig. 5 The click as the meeting of the future with the past, schematically (ASSUMPTION, the owner's word; drawn from the definitions, no run). (a) The two slits: the emitter's events spread forward through both gaps, every possibility at once (the grey fans, the future); from a realised click at a region of the screen the same line is read backward, Rule3 at -1 , through both gaps to the emitter (the dashed lines, the past); the share along the screen, the product of the two directions, alternates. (b) One gap closed by an opaque detector: its events leave the declared board; the backward reading meets one gap and the share is one gap's envelope. (c) A pair, two trees with one root: the backward reading from A's click passes the root and continues forward on B's tree (the dotted line); the root joins and nothing is signalled.

an open face and reflects; backed by a receding face the region removes from the declared board exactly what a cancelling write would, with no echo and no act. What the law gives and what the

instrument declares, precisely: what produces a click on the board is known and deterministic, the meeting, the current of the amplitudes through the region's boundary, an emitter's events bringing inflow wherever they reach a detector, a bound body at its edge when it has a current (moving, breathing or in a well's gradient), a standing symmetric body cold; what is not known is which of the possibilities becomes the measured click: when one possibility is spread over several detectors the board gives the shares and no line of the law gives, exactly, which detector fires, and the instrument draws. Exclusivity (one click per quantum, never two) and Born's rule (the credit's chance the intensity's share) are therefore the instrument's declaration and no finding of the law, for nature or against it; nature's detector efficiency and dead time are declarations of the same kind [1, what produces a click is not known]. The uncertainty relation is not a rule of its own: it is the detector's three declarations, the region of at least half a wavelength, the credit per interval and one draw per quantum, since a reading is a meeting of the two directions and a meeting is a region and an interval, never a Node and never an instant; its two readings in the paper are the which-way credit of about half the quanta and the comb's four rows at $\lambda = 8$ (ASSUMPTION, the owner's word of 2026-10-01) [3, the uncertainty relation is not a rule of its own]. A detector is no body: no read by a sense and no resonance; the source of the two experiments is a laid message, the wall a declared face, and a body is the law's name for a standing record, matter, which clicks as an instrument in the meeting form when the bodies return (Section 3.8) [3, the click ends nothing]. In the engine a detector is a region of Nodes declared in the world file, its report the `click` line of the output, the net current into the region through its front boundary Ports each interval, `reports.py`; the credit by the shares is the host's reader and never the board's act [2, the output].

3.4 The count is the record's share, and the lines that left

Every family's count is a reading of its record and no line of its own: the record's weighted share at every Node,

$$e_i = 3 \left[w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i \right] \text{div} \left(2 \sum_a p_a^2 \right) - \text{num now}_i S_6(\text{before})_i, \quad (9)$$

the conserved form's Node term over the mean of the three Link paces squared less the plain Link term, at the paces of the read, in the current's units, the share of a family's two level pairs the sum of the two; in quanta it is read over W_c , the quantum's measure of a current, by the division act, $(e_i + W_c \text{div } 2) \text{div } W_c$, the nearest whole, the unit every book, every declaration and every credit is read in. Nothing is kept at a Node beside the record and the one write's remainder: no count, no remainder of a count, no sense. Over one step of Rule3 the share at the paces the step read changes by exactly the six currents into the Node at the pair the step started from, plus the paces' anisotropy term, 0 where the three paces are equal, plus Rule3's own rounding of the level, exact in rationals from the step's three levels and its remainders:

$$e_i(\text{next}, \text{now}) - e_i(\text{now}, \text{before}) = \sum_j F_{ij} + \text{num}(\text{next}_i - \text{before}_i) \sum_a \frac{(3p_a^2 - P^2) \text{arr}_a(\text{now})_i}{P^2} - \frac{3(\text{next}_i - \text{before}_i)(r'_i - r_i)}{2P^2}, \quad (10)$$

$P^2 = \sum_a p_a^2$, arr_a the two arrivals of the axis a summed, r and r' Rule3's remainder before and after the step (THEOREM, the writer's finding checked by the Boss and the advisor and the engine's test; the count's line as it stood had read its currents one interval later, so the integer count it moved was the share one interval ahead of the record, an integer shadow with two more numbers at every Node

that no act of Rule3 read back) [1, the count is the record's share; the share's change is the currents]. So the share is the continuity of Rule3's conserved form everywhere, and over any region whose three Link paces are equal and stand, $\sum e_i$ changes by the currents through the region's boundary Links and the remainder terms alone, a telescoping sum (THEOREM) [1, the share is conserved at equal paces]. In the engine the share is one reading, `share.py`, taken from the record whenever a book, a gate or a field line asks for it, and a test asserts Eq. (10) exactly on the hand chain [2, the NodeState].

The lines that left. Until the owner's word of 2026-09-30, 23:55, the law carried two more lines at every Node, and the paper's earlier drafts were written on them: the count's line, $W_c c_{\text{next}} + r' = W_c c_{\text{now}} + \sum_j F_{ij} + r$, a count of quanta per family at every Node with a remainder, moved by the division of the current, a quantum moving whole where the remainder crossed the wall, laid once from the record's share, with the hole -1 where a total went below 0 and the click as that division at a detector's boundary Port; and the sense's line, the same division on the Wronskian's current, moving the sign. The owner's word, verbatim in translation: "throw out the count's line and the sense's line"; and in the writer's session, "the count's line and the sense's line are thrown out" (ASSUMPTION, the decision) [3, the count's line and the sense's line leave the law]. What the two lines gave, the record gives alone: the well is the form over T ; the sign is the Wronskian; what a detector sees is the current through its boundary; a body's quanta are the share of its record. The count's conservation theorem became the share's; the remainder's origin, the lay of the counts as an act, the holes and the line's own pair creation, the anti-body as a lay of -1 , the whole crossings and the velocity row of the count's centroid left with the lines, and the whole crossings the engine's tests carried agreed with the inflow a screen saw to the rounding's order, a GameBoard reading and no claim. The algebra of the exclusivity, two lines of the same day that moved the quantum itself and gave one entry per quantum at the price of the bit-exact inverse of the quantum's place, stands in the law's document under its own name and not in the law [1, the algebra of the exclusivity].

The generic Node, and everything else that left. Per family, per part, a Node holds two levels (now, before) and Rule3's remainder; per held part, one write of its sources by one division with one remainder, the second line of Eq. (1); six Ports; one line of Rule3 per part per interval, forward or backward the same line; and nothing else. The Node knows no direction of time, no pair, no root, no click and no family: at the step it holds, for one moment, before, now and next together, and D reads the three; a family is the loader's grouping of lines that share a pair, a divisor and a rest, and the readings' pairing, never known to the engine (ASSUMPTION, the owner's words of 2026-10-01: "closed for a generic implementation of a Node"; "throw them out too; via an agent that throws out everything that is not generic on a Node"; "the engine works only with families of dimension one"). With this the well's remainder, the Wronskian's remainder and the write's carries per sourcing family left the Node, the well $D_i \text{ div } T$ a reading and the sources summed in one numerator over one wall, and a structural gate in the engine's tests holds the Node to exactly this [3, the generic Node is closed for building; everything at a Node that is not the generic Node leaves; the engine works only with families of dimension one][1, the primitives, a family's write is one act].

The share under a moving well, a finding by name. The identity above holds at the paces the step read, and a step that moves the paces (a well deepening or filling under a breathing or moving body) moves the share read at the new paces beside it, with no current crossing any Link, the paces' term of the identity, which the engine's test on a breathing body shows as a swing of the quanta read over the board while a static body stays within the rounding. The owner's word: "not interested in a well"; the share stays the count, its drift under a moving well a reading of the books and no line, and the well's three lines of the paper's earlier drafts (the count is the baryon number,

the mass is the well, the mass defect is the form's fall at a kept count) are parked with the pixel, to be restated when the wells return (Section 5) [1, the paces' own change moves the share with no current].

3.5 No waves on the event board

On the GameBoard there are events, which split by Rule3 (each Node's level to its six neighbours, every possibility at once), and amplitudes, the levels Rule3 produces at the Nodes (the sum of the split events there, the record); in our world there are clicks, the detectors' reports, and nothing else. What nature calls a wave is, on the board, an emitter's events that spread (a laid message is an emitter's events declared in the file; a breathing or moving body's write is an emitter's events) and a detector that measures them or not; between the two ends there are events and amplitudes and no wave. The two slits' packet of light is the emitter's events; the fringes are where split events meet with the same sign at the screen; interference is split events meeting with opposite signs. Both directions of the meeting are these split events: forward from the emitter through the two slits the events pass both gaps, one beam that splits into two fans, every path of the future at once; read backward from the realised click by the same line, Rule3 at -1 , the split events form the past's tree, which passes both gaps back to the emitter; the click is where the two coincide, and a gap closed by an opaque detector leaves the past one gap (Section 3.2, Fig. 5). What is compared with nature is clicks with clicks; what stays on the board, the empty amplitudes, is not in our world and changes a click only where it meets another event that is realised (ASSUMPTION, the owner's word of 2026-09-30) [1, no waves on the event board][3, no waves on the event board].

3.6 The sign is the rotation sense, read and never stepped

A family of quanta carries its record as two level pairs, (re,im), two solutions of the same line stepped by one call of Rule3 with one set of coefficients, each with its own remainder; a held family's parts stay real. The charge density at a Node is the booking

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (11)$$

the Wronskian of the two pairs, which Rule3 conserves exactly with the conserved form's weights $1/p_i^2$ (THEOREM, **DM** symmetric), its local change the Link current num $(\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$ at the levels the step started from; and its total over the board is exactly invariant in the rationals under any change of the paces, since for the symmetric recurrence $\mathbf{a}_{\text{next}} = \mathbf{M}(t) \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$ the bracket $\mathbf{im}_{\text{now}}^\top \mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{now}}^\top \mathbf{M} \mathbf{im}_{\text{now}}$ is 0 for every symmetric **M** (THEOREM, the advisor's): the sign's count is the record's own, kept where the energy-like share is not, read and never stepped [1, the sign is the rotation sense; the sign's count is the record's own]. A real record ($\text{im} = 0$) has $W = 0$, so light, the sign holder's own real record, is neutral; a rotating record has W of either sign by its sense. The holder of the sign is sourced by W through its one write; who reads it is decided by the family's dimension and by one old rule, and by no name. Every family of quanta has a dimension, its declaration per family and per run, the shape of its parts: one real line, neutral matter, and light itself, the holder of the sign's own real record; or a plane of two level pairs, charged matter alone; a held row's shape is its sources' count, one real line per source (the binding holder one, gravity one with three tensions, the holder of the sign one), never a plane, and the pair family is two real lines of light never summed at a Node; the dimension is never a reading of a Node's state, so a neutral body and a charged body of one pair are two families; a family with a plane has a Wronskian and reads the holder of the sign, a family of one real line has $W = 0$ and does not, neutral by itself; and a row does not read its own level, so light, that row's own record, does not read the holder of the

sign, a static charge does not bend light, and light's own W is no charge; from the pair and the dimension the holds, the sources, the readers and the act are derived with no name (ASSUMPTION, the owner's words of 2026-10-01, "every family has a dimension") [3, every family has a dimension]. The reader's q , the sign of its own Wronskian entering its step's coefficients, left the law with the owner's word "there is no sense; throw it out; everything from the generic step of a Node": the Node reads no coefficient from its own state, so the step back reads what the step forward read and the back-in-time gate is MATCH without exception; in its place the holder of the sign declares in the universe file how it acts on its readers, the plain read of its level into the paces, or the rotation of the two-part record by the angle of its level over Γ , three integer shears with no remainder and no new state, under which a record rotating at $+\omega$ rotates at $\omega + \theta$ and one at $-\omega$ at $|\omega| - \theta$, like and unlike differing by the record's own sense; the rotation is a round of its own after Bell's first look, and the plain read by the two-pair records stands until it (ASSUMPTION, the owner's words of 2026-10-01; the engine on **main** at this writing still reads by sign in the light-only worlds, where nothing sources W) [3, there is no sense; who reads the holder of the sign is decided by the record's dimension][1, the sign holder rotates the two-part record].

The charge is the record's. For a record $(\text{re}, \text{im}) = (A \cos \omega t, eA \sin \omega t)$ the booking is $W = -eA^2 \sin \omega$: the charge is the record's size times the ellipticity e of the rotation, a property of the record that Rule3 conserves (the two levels obey one linear line, so their ratio is carried, not written). A circular quantum ($e = 1$) has $|W|/D = 1/\sin \omega$; a nearly real cloud of 1,836 quanta with $e = 1/1,836$ has the charge of one circular quantum; a body is one mode with one sense and one ellipticity at all its Nodes. What it costs, an input where nature has a measured number and no click row yet: nothing in the law quantises e , so a body's charge is continuous, and $|q_p| = |q_e|$, nature's to 10^{-21} , is an input and not derived; the line that would derive it is a hypothesis under its own name, *the second level's amplitude is a count of its own* (Section 8.4) [1, the sign is the rotation sense; the hypotheses under their own names].

3.7 Light is born by the write

Light is the sign holder's own record: the holder of the sign is one record whose time part is its real level pair, stepped by Rule3 at the paces of its reads (the content, so light is bent) and written by the hold from the Wronskian of the records that read it. A static rotating body writes a static level (its well in the sign holder, Coulomb's) and radiates nothing; a breathing or moving body writes a varying source at its Nodes, the row carries it away at one Link per interval in the vacuum, and those spreading events are the light (DERIVED). Its count is the share of the sign holder's record, read like every family's, 0 where its record is 0; a detector reads the light's current through its boundary in quanta of W_c , and the held rows' travelling events, the quanta of what nature calls a gravitational wave, enter a detector the same way, the law's prediction and no declaration, a static field seen by no detector. A body's quanta that leave it stay its family, messages of matter; no quantum changes family at a crossing, no ledger of bodies stands beside the Nodes, and a body's Nodes are where its record's share stands about its declared Nodes. A body reads the sign holder at its own Nodes, so a breathing body reads the light it writes and its pace answers, the radiation reaction, in the same loop [1, light is born by the write][3, light is born by the write].

3.8 The click ends nothing

Assumption 3 (the click ends nothing). The click is a detector's reading, local to its boundary; the record goes on after it and nothing ends anywhere. What a detector absorbed leaves the declared board through the face that backs it; the other possibilities go on splitting and leave through the

receding faces, events that were not realised, in no click. The ending of a record at a click by the detector is decided against and is no hypothesis of the law.

The owner's decision of 2026-09-30 [3, the click ends nothing], the law's claim and not a deficiency: perhaps nature is local. No signalling holds trivially, since nothing at one detector changes the amplitudes at another; the Outside reads the clicks alone. The measurement needs no collapse: the record never ends, the click is a reading, and the backward run from the click is the reading of the realised past and no write. A bound body, when the bodies return, clicks as an instrument in the meeting form: the future is light's forward record at its boundary with every possibility, the past is the body raised by one quantum read backward through its boundary to the source, and the meeting is where the two carry the same signal, which for a body is the same frequency, the resonance of its rotation with the light's, the detector's own question and no added rule; the body's receding face is its rotation, what enters at its line stays as its share and the rest leaves; the draw stays in the credit, among the bodies that could absorb one is credited the whole quantum by the shares; the emission is the same meeting with the roles reversed, and a body at rest and symmetric clicks nowhere (DERIVED, the advisor's reading at the owner's word; parked with the pixel, the absorption look of the documents, row (y), its first look) [3, a Node of the board is read only through a detector][1, the rows against nature, (y)].

3.9 The price of locality

The law's word, decided, read against nature's clicks alone [1, what is open, item 4; the rows against nature, (g) and (h)]:

- (i) *Exclusivity and Born, declared.* A quantum spread over several detectors is shares of the current at each; the law gives the shares and the instrument gives the click. A dilute packet, under one quantum per Node everywhere, rises click by click as Born by the declaration, and the earlier finding of the paper's drafts, that the pattern from single clicks does not form on a cold board, is a finding no more, for nature or for the law.
- (ii) *The which-way.* An opaque detector in one gap absorbs that gap's events whether or not it is credited, and the screen sees the other gap alone, no fringes, one gap's envelope: a closed slit by construction, in nature too; the which-way measurement proper, the path marked without absorption, is beyond the instrument the law has, and the paper says so (Section 7.2).
- (iii) *Bell.* The pair is one laid record, a common cause; its phase is a shared classical variable inside Bell's λ , and the detectors are never at one Node. A local credit of that classical phase, every quantum counted, gives $|S| \leq 2$, and the run gives 2.000 to three places by the signs on both sides, the triangle's identity; a two-port detector crediting by its contrast reached nature's cosine at the efficiency 0.64 through the detection loophole, a finding on the instrument's declaration and no longer the paper's Bell. The meeting through the pair's root (Section 3.2) is what reaches nature: each side reads one half at every port, nothing is signalled, and the blind row of the run to come, two beams of the pair family to two declared analysers, is $E = \cos 2(a - b)$ and $S = 2.83$ with every quantum counted, the share as the density of the meeting being the owner's declaration and not a derivation; the loophole-free test of 2015 that measured the sum saw 2.42 ± 0.20 [9] (Section 7.4) [3, the click is the meeting of the future with the past].

The two slits read (i) and (ii), Bell reads (iii), and their rows are the law's answer, not a line to add. The paper writes them as they stand.

3.10 The postulates

1. The world is Nodes and Events. Space is the GameBoard; an Event is a change of a NodeState, a level stepped by Rule3 with its remainder; a click is a detector's report of the currents; nothing else happens.
2. A Node holds bounded local information: its NodeState, fixed in size, read from itself and its six neighbours; no read beyond them, no list that grows with the world, nothing of the past kept. A self-field estimated or subtracted from anything global is forbidden, whatever the size of its final answer.
3. Consistency is local and causal: an Event first changes its own Node and travels Link by Link, one Link per interval, each Node updating on receipt under the same rule; the one exception is the production of a click, the instrument's credit of one quantum to one detector by the shares over every declared detector, the one non-local place, not known and not derived; and no quantum changes family at a crossing.
4. The causal speed is one Link per interval, built in; every other speed is a rational, a count of Links over a count of intervals.
5. Every physical calculation is on bounded integers; there is no float, no root and no draw in the law; a Node keeps only the law's own numbers: each part's two levels and Rule3's remainder, a held part's one write remainder, the family's pair and the Node clock from the content there, and nothing else; its count, its sense, its currents and its tension are readings.
6. A measurement is a detector's click, a reading of the state: the net current into the declared region through its front boundary Ports over one interval, reported with the region and the family and never a Node; one quantum is credited to one detector by the shares, the instrument's declaration, and which meeting becomes the click is not known and not claimed. Only a click is compared with nature or pinned as an expectation; displays and the host's readings read state and write nothing.
7. The Inside is the GameBoard, where no one measures, deterministic and bit-reversible; the Outside is the detectors and their clicks, realised events, the only thing claimed to represent nature.

[1, the postulates]

4 The families from the rule

A family is a band of the one rule and not a declaration, and its row holds nothing the rule and the geometry fix: its pair $[\text{num}, \text{den}]$, $\cos \omega_0 = \text{num}/\text{den}$, and what it holds, the content or the sign, at a divisor. Every pair is written over the GameBoard's one denominator Γ , so the file holds the numerator m alone, $\cos \omega_0 = m/\Gamma$, and the wall is $w = 6\Gamma^3$ for every family; a reduction to lowest terms is a reading and never the loader's act, so every family's remainder has the one resolution [1, the families from the rule; the mass is one integer; the wall is one].

How a family enters. A family enters the world as one row of the universe file with two keys and nothing else: its pair $[\text{num}, \text{den}]$, and, for a held family, what it holds, the content or the sign, at its divisor E_s ; a family of quanta holds nothing. The rest is derived from the two keys and declared by no row: its rotation at rest, $\cos \omega_0 = \text{num}/\text{den}$, and its band; its coefficients and its wall, $w = 6\text{den}\Gamma^2$ (Eq. (1)); its parts, the time part alone but for the massless row that holds the content, which carries three tensions beside it (Section 4.3); its reach, set by its gap (Section 4.4); its reads, every family of quanta reading every holder of the content at the weight 1, and a record laid as a plane of two level pairs reading the holder of the sign as well; its writes, since a family sources what

it reads at the weight it reads with; and its clicks, which every record makes, a holder's included, since no row declares whether it clicks. A new family is one more row and no code (Section 9). Table 1 gives the rows of the universe, and Table 2 the numbers of the world beside them [1, a family's declaration; the families from the rule][2, how to add a family].

4.1 The exact bands, and the third closed as absent

The line of one Node at rest, $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$, is exact in integers only where $2 \cos \omega_0$ is an integer, so the rule carries exactly three exact rotations, $2 \cos \omega_0 = 1, 0$ and -1 , the periods 6, 4 and 3, the pairs $[1, 2]$, $[0, 1]$ and $[-1, 2]$ (THEOREM). A record on an exact band at rest at the vacuum's pace leaves no remainder in its rest rotation, so it never clicks and only turns: no remainder, no click. The exact bands are exact only in the vacuum: at a Node of pace p_0 the band's rest rotation is $2 \cos \omega = 2 - 2(\text{den} - \text{num})(p_0/\Gamma)^2$, an integer at $p_0 = \Gamma$ alone, so a record on an exact band inside a well leaves a remainder and clicks like every family's [1, the exact bands; the exact bands are exact only in the vacuum].

The third, $[-1, 2]$, is a band the rule allows and the universe does not use, the mirror of $[1, 2]$: one family on it holds no standing body (its read is negative, so its own well pushes its rest rotation up into the vacuum's band and the record leaks). The three colours are the three axes of a click's Ports, a detector's reading of the click's axis, no charge of the body and no force of their own; whiteness is their balance at rest. So the quarks are not in the law: the white body, the tube, the charges $2/3$ and $1/3$, the generations, up and down leave with the third, and the band $[0, 1]$ of the period 4 is the same, a band allowed and unused [1, the third and the colours, closed as absent][3, the colours are the three axes].

4.2 The rule's own universe

The universe is the one whose every number is the rule's: the pairs from the integers 1, 2 and 3 of the three exact rotations and the three axes, every divisor 1 but gravity's, and the free numbers named. Its rows stand in Table 1, as the file `examples/events/rule.json` holds them and as the rule derives the rest from the two keys of a row, its pair and what it holds (DERIVED) [1, the families from the rule; the rule's own universe][2, the loader and the files].

The universe's integers: the Node clock $\Gamma = 6,000$, a multiple of 6 so that every pair is exact over it; the quantum's action $T = 2^{15}$; the integers' width 63 bits, a bound and no physics; and the vacuum content, the massless row's rest, 0 by nature's clicks (Section 4.6). Every family of quanta reads every holder of the content at the weight 1 and sources what it reads at the same weight; a record laid as a plane of two level pairs also reads the holder of the sign, and one laid as one real line does not (Section 3.6); matter reads gravity and the binding holder plainly, $x = (L_b + L_g)/\Gamma$, and the bound charge; light reads the same holders of the content and is bent by them, and not the holder of the sign, its own row. The three of Rule3 is the three axes; the three axes give the three tensions the massless row carries beside its time part, $1 + 3$, the parts Rule3 reads and no other; every other family is a scalar. So the files hold Γ , T , the matter pair's m , the binding holder's pair, gravity's E_g and the vacuum content, every other divisor 1, and nothing else of physics; Γ and T are units, as c and $\hbar$ are in nature, and no line of Rule3 fixes T , the rule being linear and its form's scale free. The crack, named: a second massive scalar family, a second mass, the rule does not forbid, and nature's spectrum of masses is the open row under this line [1, the final unification][3, two units and the bounds; the standing goal].

Every row of the file is a record stepped by Rule3 at its pair and sourced by the hold's division act, the vacuum's row and every holder with a gap alike; no laid row, no fixed point computed

Table 1 The rule’s own universe, its five families as the file `examples/events/rule.json` holds them (DERIVED from the two keys of a row, its pair and what it holds): each family with its rank, its pair [num, den] (in brackets, written over the Node clock $\Gamma = 6,000$), what it holds at its divisor, its parts and its free number.

Family, its rank	Pair	Holds, at the divisor	Parts	Its free number
gravity, the time part and the three tensions	[1, 1] ([6000, 6000])	the content, E_g (at least 10^6 at $\Gamma = 6,000$, a click bound from the two slits’ screen; 1,000 in the parked looks)	1 + 3	E_g , the hierarchy; its rest c_{vac} , 0 by nature’s clicks
the binding holder, the strong force; its own events have the gap $\cos \omega = \text{num}_b / \text{den}_b$	[2400, 2401]	the content, 1	1	the pair, its range $R = 20$ Links, tied to Γ by the proton
the charge, whose record’s events are light, born by the write; read by the two-pair records and not by its own	[1, 1]	the sign, 1	1	none
the bound charge (the polarisation), the exact band	[1, 2] ([3000, 6000])	the content, 1	1	none; a field of under one Link’s reach
matter, the scalar; a quantum that leaves a body a message of matter	[m , Γ] ([4000, 6000])	nothing	1	m , the mass

outside Rule3 during the run and no act beside the six verbs and the division act remains in the law or the engine; the start’s rest is the fixed point of the division act with its remainder, the state Rule3 holds static, computed once by the generator (ASSUMPTION, the owner’s word) [3, nothing that is not Rule3 remains].

4.3 Scalar, plane, and one plus three: what each family carries, and how gravity enters as a clock

What a family carries is not chosen by a name but by what Rule3 reads and what a record can write (DERIVED, the owner’s words of 2026-10-01 on the dimension; Section 3.6). Rule3 reads two things of a Node and no third: the clock’s pace p_0 , one number, and the Link’s paces (p_x, p_y, p_z) , the three directions of the vector the rule works with; so a held row that acts on the paces can carry at most one time part and three axis parts, 1 + 3, and nothing a read would not close. A record of one real line, neutral matter and light, has a form D and a stress G_{aa} and no Wronskian; a record of a plane, two level pairs, charged matter, has a Wronskian W too, the sign; a held row is always real, one line per source, since a plane’s own W would be its own source. From this the shapes follow with no choice: *the scalar*, one real line, for matter and for light, the holder of the sign’s own record, and for the binding holder and the bound charge, sourced by the form alone; *the plane*, two real lines in one plane, for charged matter alone; *one plus three*, the time part and the three tensions T_{xx} , T_{yy} ,

T_{zz} , for gravity, sourced by the form and by the three diagonal stresses, the diagonal of the record's own stress tensor (Section 2.7); and *two real lines never summed at a Node* for the pair family of light (Section 3.2). What the law does not take: a vector part of the held rows (a current's read, frame dragging) and the off-diagonal parts of the stress, which Rule3 reads nowhere, so a write of them would close no read; magnetism needed none of them (Section 6, (b2)), and their absence is the one finding on gravitational waves' polarisations (Section 8.4, item 23) [1, the final unification; the tension][3, every family has a dimension; the held rows, and the hold in the count's words].

How gravity enters: a clock that is slowed, and no space that is curved. There is no metric, no force and no law of motion in the engine. Gravity enters in three steps, each a line of this paper. (i) *The write*. At every Node, each interval, every family of quanta writes its form over the action, $D_i \text{ div } T$, into gravity's row over the divisor E_g , the second line of Eq. (1); a static record writes a static level, a breathing one a varying one (Section 2.8). (ii) *The row*. Gravity's row, $[1, 1]$, steps by Rule3 at the pace 1, so around a static source it rests at Poisson's, $3G(r)s$, $3s/(4\pi r)$ far away, and a change travels at one Link per interval, retarded, with the exact inverse kept; the three tensions step beside the time part (Sections 2.2 and 8.1). (iii) *The read*. Every family reads the row's level c at its Node into its own coefficients, Eq. (5): the clock's pace once, $p_0^2 = (\Gamma - c)^2 + c^2$, and each Link's pace twice, $p_a = \Gamma - 2c - t_a$, with the axis's tension t_a on it. That is the whole of gravity. A slower clock is what a deeper well is: every massive band's rest rotation slows by the same factor $(p_0/\Gamma)^2$, so two bound bodies at one level shift their clocks alike and a far detector reads the emitter's slow clock, the redshift, and bodies fall alike, the equivalence principle for the bodies of nature (Section 6, (a) and (h)). What nature calls curved space is here the Link's pace: along every Link through the well a record's speed falls by the level twice, light's index is $n = 1/(1 - 2x)$, and a beam bends by $4U_b$, twice the clock's share, while an orbit's periapsis advances by $6\pi U_p$ once the clock carries its second order, the composition of the slowing unit by unit (Section 2.5; Eq. (13)); the tensions add the stress, so a moving record and light's own pressure deepen the content around them as nature's do. No space is bent: the Nodes stand where they stand, the Links are what they are, and only the paces at which the rule divides are lower where the well is deeper, once for the clock and twice for the Link. In the engine this is the signed read of the paces from the held rows' levels at the interval's start and the hold's one write at its end, `features/signed_read` and `features/hold`, and the held row's own step by the same Rule3 [1, the paces; the clock once and the Link twice; the constants][3, the Node clock and the ruler].

4.4 The reach of a held family is its pair

A held family's rest solves $\text{num } \Delta^2 a - 6(\text{den} - \text{num}) a = -3 \text{den } \sigma$ (its line with the source once an interval, Δ^2 the six-neighbour second difference), so outside a body its rest falls by $e^{-\kappa}$ per Link with $\cosh \kappa = 3 \text{den}/\text{num} - 2$, the family's own band at zero rotation: the gap is the mass of the records and the inverse reach of the field, one number (DERIVED). $[1, 1]$ has no gap and its rest is Poisson's over the whole GameBoard; $[1, 2]$ has $\cosh \kappa = 4$ and falls by 0.127 per Link, a contact well of 0.41 Links' reach; and a pair with num just under den has $\kappa^2 = 6 \text{den}/\text{num} - 6$ to the second order, a range $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$ Links [1, the well].

4.5 The two rows: the binding holder and gravity under a divisor

Assumption 4 (the two rows, decided with the caveat). The well of a record, $D_i \text{ div } T$, is laid into two holders. The binding holder, a family with a gapped pair $[\text{num}_b, \text{den}_b]$, $\text{num}_b < \text{den}_b$, at the divisor 1, stepped by Rule3 at its pair as the vacuum's row is and started at its rest, which with the gap is a sum with a range, $R = \sqrt{\text{num}_b/(6(\text{den}_b - \text{num}_b))}$ Links and $e^{-r/R}/r$ beyond it: it binds

the bodies, it is the law's strong force; its own waves have the gap $\cos\omega = \text{num}_b/\text{den}_b$, a quantum with a mass whose Compton length is R , Yukawa's own line [10] and no input. The massless holder [1, 1] at the divisor E_g declared by the file: gravity, long-ranged and weak by E_g . A family reads both levels into its pace at the weight 1.

The first pair is [2400, 2401], $R = 20$ Links, $m^2 = 0.0025$, the quantum's rest frequency 0.029 per interval, the pair tied to Γ by the proton: a cloud of 1,836 quanta is bound under $R = 20$ and not under $R = 10$ or 15 (Section 5.3). In the looks E_g is the run's own number, large enough that gravity does not enter a body's binding (1,000 suffices at the pixel, 903/1,000 under one level); nature's E_g is about 3×10^{42} , 10^{36} times $1,836^2$, and the hierarchy is the file's one number, as G is nature's. The reservation, in the owner's words, we may have been wrong: the line stands as a decision with a reservation and reopens on any of (i) the three tests, (ii) the bodies not standing under the screened sum, (iii) the charged cloud look reading electric over gravitational between two bodies as $E_g/1,836^2$ and nature's other way, (iv) the finding of scale below, which this line does not touch and which may be the deeper error [3, the two rows][1, what is open, item 20].

The road not taken, kept as the law's word. Under every divisor 1 the force that binds a body and the force between bodies are one row, the well read at the weight 1; the gravity between two quanta is then of order one against nature's 5.9×10^{-39} for the proton, and the electric force between two protons, a charged cloud being a nearly real record with one unit of W among 1,836 quanta (Section 3.6), is $1/1,836^2$ of the gravity between them, where nature's is 10^{36} times larger. That is the price of every divisor 1 with its number, written as the law's word on its own universe, a computed number set beside nature's and no click [1, the rows against nature, (v)]. *The reading of scale*, the same statement with no number: matter's rotation at rest, $\cos\omega_0 = 2/3$, fixes the electron's reduced Compton length in the lattice's units, light's speed over the rotation, $0.577/0.841 = 0.69$ Links; nature's proton is $1/400$ of that, 0.0015 Links, while the law's cloud of 1,836 quanta has the radius $2\Gamma/C$, at least 6.5 Links at $\Gamma = 6,000$ and larger at every larger Γ : nature's proton is bound thousands of times smaller than the well binds anything, so the well row binds too strongly at a distance and too weakly up close to be both the proton's binding and gravity [1, the rows against nature, (v) (2)].

4.6 The vacuum content

Assumption 5 (the vacuum content is 0, a declaration fixed by nature's clicks). The massless row rests at a declared level everywhere, the vacuum content, the key **rest** of the universe file, derived by no line of the law; nature's clicks fix it at 0 in the file's integers at $\Gamma = 6,000$.

A constant is the massless row's own rest and its step keeps it ($a_{\text{next}} = S_6 \text{ div } 3 - a_{\text{before}}$ returns it); every reader's paces are $\Gamma - 2c_{\text{vac}}$, the same physics at an effective Γ , so the law cannot tell 60 from 0 and only nature's clicks can. The clicks that fixed it, the first row of the calibration series (Section 8.5): light reads the vacuum content into its paces at the speed $(1 - 2x)/\sqrt{3}$ while the massless row's own events step at $1/\sqrt{3}$, so the two arrive apart by $2xL\sqrt{3}$ intervals over L Links at long wavelength, and by 17.9 intervals over 458 Links at $x = 0.01$ and the wave number $\pi/4$ from the two bands exactly (COMPUTED, Section 8.5); nature's GW170817, the gravitational wave and the gamma-ray burst 1.7 seconds apart after 1.3×10^8 years, bounds the difference of the two speeds between -3×10^{-15} and $+7 \times 10^{-16}$ of light's [11], so the law's $2x$, gravity's lead over light, is at most 7×10^{-16} , c_{vac}/Γ is below 10^{-15} and the file's integer is 0: the law's cosmological constant is 0 to every measurement's precision, the first number of the series entered as clicks. At the rest 0 light reads only the bodies' wells and the field is non-negative wherever a well stands. A holder

with a gap sourced by a changing well rings with its own travelling events, the law's mesons, whose troughs below 0 far from the body light refuses, rightly, as a real runaway of Rule3 (Section 2.5): that is the reason for light's refusal below 0, and no reason for a rest above it. The two experiments of Section 7 read no rest, since a fringe's spacing and a calibrated phase read none [1, what is open, item 22][3, the vacuum content is 0, a declaration fixed by nature's clicks].

4.7 The world's numbers, and how the universe was chosen

The universe is chosen by one principle stated before any row: one rule closes everything; the families are its inputs and no part of it, each a band of the rule named by its pair; the units are a choice of scale; and the universe holds as few free numbers as the law allows (ASSUMPTION) [3, the standing goal; not one number in the engine, and as few in the files as the law allows]. The engine holds no number of physics, so every number of the world is a key of its universe file, and choosing the universe is choosing which keys stand there and why. The road to them, step by step:

- (i) *A family is a band of the rule*, not a declaration: its row holds only what the rule and the geometry do not fix, its pair and what it holds (the paragraph on how a family enters, above).
- (ii) *Three exact rotations*. The rule carries exactly three, the pairs $[1, 2]$, $[0, 1]$ and $[-1, 2]$ (THEOREM, Section 4.1).
- (iii) *The three of Rule3 is the three axes*. The massless holder of the content carries its time part and three tensions, $1 + 3$, and every other family is a scalar, so the parts carry no free number (Section 4.3).
- (iv) *One denominator and one wall*. Every pair the universe fixes is written over Γ , so a mass is one integer m , and Γ is a multiple of 6 so that every such pair is exact over it.
- (v) *The pairs from 1, 2 and 3*, the integers of the exact rotations and of the three axes (ASSUMPTION): $[1, 1]$ for the two fields, the holder of the content, gravity, and the holder of the sign, the charge, whose quanta are light; $[1, 2]$ for the bound charge; and $[2, 3]$, $\cos \omega_0 = 2/3$, the first pair past the exact bands, for matter. The bands $[-1, 2]$ and $[0, 1]$ are allowed and unused, since a family on $[-1, 2]$ holds no standing body (Section 4.1).
- (vi) *Every divisor 1*. With every divisor 1 the universe carried no free number but the mass: the rule's own universe in its first form.
- (vii) *Two numbers where nature's scale forced them*. Under every divisor 1 the force that binds a body and the force between bodies are one row, so gravity is as strong as the binding, where nature's gravity is weaker than the electric force by about 10^{36} (Section 4.5); so the law took two numbers and no more, the binding holder's gapped pair, the strong force, and gravity's divisor E_g , the hierarchy (ASSUMPTION, decided with the reservation stated there).
- (viii) *The vacuum content by clicks*. The massless row's rest is the one number the law cannot read from inside, since a reader's paces at a rest c are $\Gamma - 2c$, the same physics at an effective Γ ; nature's clicks fix it at 0 (Section 4.6).

What remains stands in Table 2: two units, Γ and T , as c and $\hbar$ are nature's; one bound, the integers' width, which is no physics; and four numbers of physics, matter's numerator m , the binding holder's pair, gravity's divisor E_g and the vacuum content, every other divisor being 1. The two experiments' worlds read a light-only copy of the file with the three holders of the content at the divisor 10^6 , so that light's own well stays under one level. That this universe is the universe is a decision and not a theorem (ASSUMPTION): the rule's integers fix every row, and the law names where another universe could stand. The same integers give a second massive pair, $[1, 3]$, a second mass the rule does not forbid and the universe does not use, the crack, under which nature's spectrum of masses is the open row; and the numbers that would close the universe toward nature, the fine-structure constant, the electron's mass over Planck's and the proton's over the electron's,

Table 2 The numbers of the world, each a key of the universe file `examples/events/rule.json` and none of the engine: its value in the file, its kind and what fixes it.

The number	In the file	Its kind	What fixes it
the Node clock Γ	6,000	a unit, as c is nature's	a multiple of 6, so that every pair of the universe is exact over it
the quantum's action T	2^{15}	a unit, as $\hbar$ is nature's	declared once; no line of Rule3 fixes it, the rule being linear
the integers' width	63 bits	a bound, no physics	the machine's integers
matter's numerator m	4,000, so $\cos \omega_0 = 2/3$	the mass	the pairs from 1, 2 and 3: [2, 3], the first pair past the exact bands
the binding holder's pair	[2400, 2401]	the strong force, its range $R = 20$ Links	tied to Γ by the proton: a cloud of 1,836 quanta binds under it and not under a shorter range
gravity's divisor E_g	1,000 for the bodies' worlds, 10^6 for the light-only worlds	the hierarchy, as G is nature's	bounded below by light's own fringes, at least 10^6 at $\Gamma = 6,000$ (Section 8.5); nature's about 3×10^{42}
the vacuum content c_{vac}	0	the cosmological constant	nature's clicks, GW170817 (Section 4.6)
every other divisor	1	none	the rule's own universe

are open by name and not derived (Section 8.4) [1, the rule's own universe; the final unification; the tree][3, the rule's own universe is the universe; the final unification].

5 The bodies

5.1 What a body is

A body is its record over its Nodes and the quanta that record carries, and never one Node: every body is many Nodes, and nothing gathers it. Its count, the number of its quanta, is the share of its record read in the vacuum's coefficients, $e_i = 3 \text{ den}(\text{now}_i^2 + \text{before}_i^2) - \text{num now}_i S_6(\text{before})_i$ over the wall W_c , summed over its Nodes, a reading and no state of the board (Section 3.4); its form $D_i \text{ div } T$ enters Rule3 through the pace alone, $x = (L_b + L_g)/\Gamma$. The record of the body's family splits by the read and is bound by the wells its counts source: by the binding holder's screened sum with the range R , while gravity's $1/r$ is weak by E_g and enters no body's binding [1, what a body is]. No law is written for a body: the one rule is written for the Node, and a body is the support of a mode of that rule; every proposal is a row at a Node or it leaves or is derived. A body has no law of motion of its own: its record moves by Rule3 alone, the count follows by the count's line, the momentum is a reading of the record's current, and in a pace gradient its packet accelerates as the mode's top velocity times its redshift per level times the level's difference across it [3, what a body is].

The four lines of the body (DERIVED from the line, with the decisions named): (a) *the body is the fixed point of its binding row*: its form sources every held row its family sources, its record is the standing record Rule3 makes in those paces, its count the weighted share that record lays, and it stands where the sources return themselves; (b) *the form sources the fields*: a held row's level is the rest the start solves from the bodies' forms at the row's divisor, then Rule3 steps it; (c) *the quanta follow the record*: a body's quanta are its record's share, carried wherever its record's current carries it, the well $D_i \text{ div } T$ each interval being the record's quanta as the fields' source; (d) *the fall*, for a free packet and for a cloud, $a = \Delta |d\omega_b/dc| \nabla c$, Δ the band's curvature at $k = 0$ and $d\omega_b/dc$ from

Rule3's coefficients at the body's pace and rotation; a compact pixel falls by its tail alone, slower, so this line does not hold there. The world file names the body's family, its Nodes with their counts and the quanta of other families it holds, and nothing else: no momentum, no spin, no emitter, no sign, no seed, no stop, no period; its two levels are the generator's mode file beside the world [1, the four lines of the body].

5.2 The functional and the dilation

Let a body of M quanta have the record ϕ over its region, $\sum \phi_i^2 = 1$, its counts the form $c_i = M\phi_i^2$, and let every held row r its sources be a hollow ($\sigma_r = +1$) or a hill ($\sigma_r = -1$) with the static kernel $\mathbf{G}_r$ of its line, $(\Delta^2 - \kappa_r^2)a = -3 \text{den } \sigma / \text{num}$ ($\kappa = 0$ for the massless row), and the divisor E_r , so that the well at a Node is $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$. To first order in Φ/Γ the well line raises the one-Node rotation $2 \cos \omega$ by $4g\Phi/\Gamma$, $g = (\text{den} - \text{num})/\text{den}$, and there the fixed point of the binding row is a critical point on the sphere of

$$F[\phi] = \frac{\text{num}}{3 \text{den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2, \quad (12)$$

the band's pressure less the wells' energy, each well counted once for the record being its own source (Hartree's form; THEOREM: the record is the top mode of Rule3's symmetric form at the paces, $\mathbf{M}(\Phi) = \mathbf{M}_0 + (4g/\Gamma) \text{diag}(\Phi)$ at first order, and $dF/d\phi = 2\lambda\phi$ on the sphere is the fixed point's own equation with $2 \cos \omega_b = 2 \cos \omega_0 - \lambda$). Let $\phi_R(x) = R^{-d/2} \phi(x/R)$ be the body dilated, d the GameBoard's dimension; the pressure is K/R^2 and each row's well energy W_r/R^{s_r} , s_r the row's scaling power at the body's size: $s_r = d$ beyond the row's reach (a contact), $s_r = d - 2$ within the reach or for the massless row (Poisson's $3/(4\pi r)$ in a box) [1, the functional and the dilation].

Theorem 4 (the stable body). *At a fixed point of the binding row of width R the virial $2K = \sum_r \sigma_r s_r W_r$ holds, and the second variation of F along the dilation is $R^2 F''(R) = \sum_r \sigma_r s_r (2 - s_r) W_r$. A body is stable only where this sum is positive; it is a minimum of F where further the Hessian on the modes orthogonal to the dilation and the translations is positive.*

Proof. $F(R) = K/R^2 - \sum_r \sigma_r W_r R^{-s_r}$; $F'(1) = 0$ is $-2K + \sum_r \sigma_r s_r W_r = 0$; $F''(1) = 6K - \sum_r \sigma_r s_r (s_r + 1) W_r = \sum_r \sigma_r (3s_r - s_r^2 - s_r) W_r$: Derrick's scaling [12], Pohozaev's identity [13] on the GameBoard [1, the functional and the dilation]. $\square$

The signs: $s(2 - s)$ is $+1$ at $s = 1$, 0 at $s = 2$, -3 at $s = 3$: a hollow holds the body where its well falls slower than the pressure and breaks it where it falls faster, and a hill does the opposite. From them: (a) one hollow beyond its reach binds every mass on a chain, is flat along the dilation on a plane, and in a box has one critical point, a maximum along the dilation, a saddle for every mass: the body disperses beyond it and collapses within it; (b) a massless row holds the body: Poisson's well in a box has $s = 1$, $F = a/R^2 - c/R$ has the one critical point $R = 2a/c$, a minimum for every mass (Pekar's body), its width falling as E_g/M ; beside it a hollow beyond its reach adds a barrier, $F = a/R^2 - b/R^3 - c/R$ on a Gaussian body, the critical points $R = (a \pm \sqrt{a^2 - 3bc})/c$, the wide one the minimum and the narrow one the saddle, none where $3bc > a^2$, the collapse; the largest mass with a minimum is $M_{\max} = a\sqrt{E_b E_g}/\sqrt{3b_1 c_1}$ (COMPUTED, the script beside the paper: on the matter pair [4000, 6000] with a binding row of reach 2.02 Links at $\Gamma = 6,000$, $M_{\max} = 4,450\sqrt{E_b E_g}$); (c) the horizon ends a collapse: beyond first order the well line saturates and the Link's pace weakens the pressure, both vanishing together at the pace's floor, so the collapse ends where a Link's pace reaches 0, a frozen cluster of horizon Nodes and no body of ordinary matter; (d) the hill, a hypothesis under its own name, *the hill holds the body*, and (e) *the cooling*, the hypothesis that a

Table 3 The cloud, the wide branch of the binding row’s fixed point, along its branch (DERIVED from the fixed point, a reading of the algebra and no click): the total count, the rotation $2 \cos \omega_b$ against the band’s top 1.3333, the rms radius in Links, the count at the centre Node, the well and the inertia over the count and over each other, and the fall against a free packet’s. An open 41-cube at $\Gamma = 6,000$, matter [4000, 6000], $T = 2^{15}$, under the massless row alone; the counts in the vacuum’s share at the Node, which for a cloud is the engine’s count to a percent.

Total count	$2 \cos \omega_b$ (top 1.3333)	rms radius	Centre Node	Well / count	Inertia / count	Inertia / well	Fall / free
4,000	1.3509	8.7	2.6	0.985	1.061	1.077	0.92
3,000	1.3437	10.3	1.0	0.991	1.037	1.047	0.95
2,000	1.3383	11.5	0.4	0.995	1.022	1.027	0.97
1,500	1.3359	12.0	0.3	0.996	1.016	1.020	0.98

body above its minimum sheds its excess as quanta of its own family and radiates light while it breathes, a cold body being one that has finished shedding. What is missing for the theorem to close: the Hessian beyond the dilation for a set of rows, the functional beyond first order, and the finite GameBoard [1, the functional and the dilation].

The surplus leaves; the mass defect, parked. At a Node of a body’s record the local rotation is read from its three levels, $2 \cos \omega_i = (\text{next}_i + \text{before}_i) / \text{now}_i$, against the local band’s top; at a fixed point every Node is bound, and a cloud not at its fixed point has a free part, the Nodes whose rotation lies in the band, which leaves at its band’s group velocity by Rule3 as it stands, its quanta with it. The bound surplus does not leave by itself, since a bound mode is stationary and Rule3 reversible (COMPUTED, the script beside the paper: a cloud twice too wide in a static well breathes about its minimum for 6,000 intervals with no settling), so evaporation sheds the free part alone. The mass defect is the form’s fall at a kept count (Section 2.6): a standing mode in a well deepening from 300 to 600 at $\Gamma = 6,000$ loses 4.0 percent of its form with its count kept, so a body that binds is lighter in every field it sources by its binding and no lighter in quanta; gravity’s source is the well and the count is the baryon number. The three lines, the count as the baryon number, the mass as the well and the mass defect as the form’s fall at a kept count, were written on the count’s line and are parked with the pixel at the owner’s word of 2026-10-01 (“the well is not of interest”), the share’s drift under a moving well standing beside them by name (Section 3.4), to be restated when the wells return [1, the surplus leaves; the mass defect is the form’s fall at a kept count; what is open, item 11].

5.3 The two branches: the cloud and the compact pixel

The binding row’s fixed point has two branches at every count above the edge, and the body of nature is the wide one (DERIVED from the fixed point, computed by the advisor; the reading the owner’s decision) [1, the two branches][3, the body of nature is the cloud].

The cloud, the body of nature. The wide branch, the continuum’s Pekar body [14]: the count spread over many Nodes, the radius about $2\Gamma/C$ Links for a count C . Its three masses, the well it sources (the sum over its Nodes of $D_i \text{ div } T$, gravity’s source), the count it resists with (the inertia, the momentum reading $n = 3Q j_a \text{ div } T$) and the weight it falls with (the momentum identity’s force term under a content gradient), agree to the order $(C/\Gamma)^2$, and it falls as a free quantum falls. Table 3 gives them along its branch on an open 41-cube at $\Gamma = 6,000$, matter [4000, 6000], $T = 2^{15}$, under the massless row alone (the tension left out as of the order $(C/\Gamma)^2$ for a wide body), in the advisor’s unit, the vacuum’s share, which for a cloud is the engine’s weighted share to a percent [1, the rows against nature, (u)].

Table 4 The compact pixel, the narrow branch and collapsed matter, from the edge to near the horizon (DERIVED from the fixed point with the tension, a reading of the algebra and no click): the count at the Node, the total count, the well it sources and the inertia, each also over the count, and the fall against a free packet's; an empty cell is a number the law's document does not give. The counts in the vacuum's share at the Node; the engine's count is about three times them along this branch.

Count at the Node	Total count	The well	Well / count	The inertia	Inertia / count	Fall / free
1,650 (the edge)	2,541	1,769	0.70	2,642	1.04	
2,000	2,640	1,637	0.62	2,621	0.99	0.27
2,500	3,005	1,636	0.54			
3,000	3,449	1,692	0.49	3,105	0.90	0.11
4,000	4,413	1,860	0.42			
4,500	4,917	1,967	0.40	4,069	0.83	0.03
5,000 (near the horizon)	5,440	2,114	0.39	4,517	0.83	

The two lighter clouds lean on the box's faces, so their true radii are larger and their numbers nearer 1 than Table 3 shows. So Galileo and Eötvös [15] hold in the law for wide bodies to $(C/\Gamma)^2$, the better the larger Γ is against the body's count. In the integer engine a cloud is sparse: at 1,500 in all there are 0.3 quanta at the densest Node, so the lay writes a quantum at a few hundred Nodes and none elsewhere, and the record shared by ten thousand Nodes carries them.

The compact pixel, collapsed matter. The narrow branch: one Node and its tail, a standing record above the top of its family's band in the wells its own form lays, its count the sum over its Nodes, at one Node near the horizon and over its tail below it; a cluster is pixels within tail reach that click at one another. Its three masses part by the binding, since the gradient reaches its quanta only through its tail, where its own well has taken most of the Link pace $((1 - 2x_0)^2 = 0.14$ at 2,000). Table 4 gives them along the narrow branch under the massless row with the tension (the counts in the advisor's unit; the engine's count about three times them, 3.7 at the Node and 3.1 in all at 2,000) [1, the rows against nature, (u)].

A free quantum of matter at rest has a well equal to its count (a plane wave at $k = 0$ carries per Node the count $b^2(1 - (\text{num}/\text{den})\cos\omega_0)/T$ and the well $b^2\sin^2\omega_0/T$, equal at $\cos\omega_0 = \text{num}/\text{den}$), and a compact pixel's well is under its count everywhere in the window, 0.70 at the edge to 0.39 at the horizon: the binding lightens, the more so the deeper. Its inertia lies near its count and not its well, and its fall is suppressed, none at the horizon: a compact body sources gravity with its well, resists with its count and falls with its tail, the three masses parting by the binding. The horizon is a horizon, the pixel at the horizon a collapsed body and no proton, and gravity through the pace is not a metric for a compact body: a metric moves a body and its binding together, here the gradient moves the record and leaves its wells to follow, and the record is sheltered in them. That is the law's word on collapsed matter and no finding against nature, since nature's bodies are wide (Fig. 6). The well has a minimum along the branch, 1,636 at about 3,000 in all, so accretion pays all the way to the horizon, where it ends. Between the two branches stands the saddle, where the total count falls with the binding (the Vakhitov and Kolokolov reading [16], $dC/d\lambda < 0$ with $\lambda = 2\cos\omega_b$) [1, the three masses of a body; the two branches].

The electron and the proton. The electron is one quantum of the matter family with the sense -1 , a message and no pixel, stable by the integer (one quantum does not disperse into fractions), its charge the proton's exactly since the sense is a sign, the positron the same quantum in the sense $+1$; it needs no band of its own. The proton is a cloud of 1,836 quanta, so $m_p/m_e = 1,836$ is the count's ratio directly, and it fixes no Γ : Γ is free and large against 1,836, as nature's Link is small

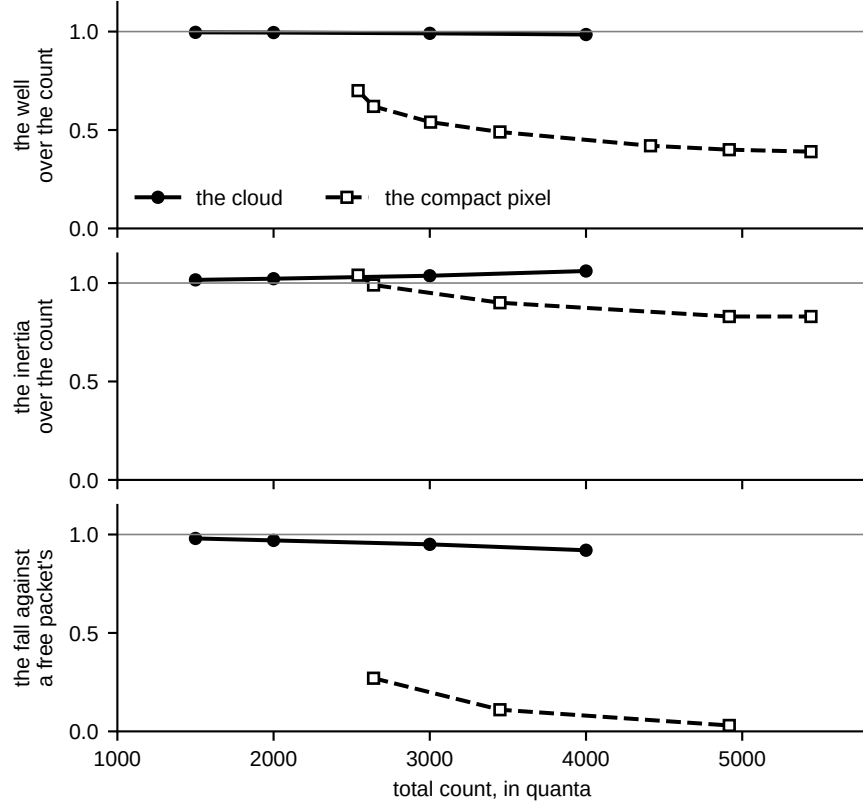


Fig. 6 The three masses of a body along its two branches, from the documents' rows (DERIVED from the fixed point, a reading of the algebra and no click; the counts in the vacuum's share at the Node), top to bottom: the well over the count, the inertia over the count and the fall against a free packet's, for the cloud (the wide branch, filled marks) and for the compact pixel (the narrow branch, open marks), each against its total count (Tables 3 and 4). For the cloud the three go to one and the fall to a free quantum's the wider it is; for the compact pixel they part by the binding. Drawn from the rows of the law's document; no run.

against a proton. And 1,836 is not derived: the law has one scale, Γ , and pure numbers of order one (the pair $2/3$, π , the lattice Green's function), so every count it derives is a fraction of Γ or a small number, and a ratio of 1,836 between two bodies needs Γ or a large pure number the law lacks; the roads tried (the collapse count, about 0.5Γ ; the lightest cloud that binds at all, about $\sqrt{8.4\Gamma}$; their ratio; the coincidence $6\pi^5 = 1,836.1$) give none. So the proton's count is an input, an integer the world file declares as every body's count is, and the law fixes the electron's, one (ASSUMPTION, the owner's decision on the advisor's computation) [1, the electron needs no band of its own; the rows against nature, (t)].

The second family's study, and the proton's size. Two lines of Rule3 decide the road of a second, heavy family of matter before any fixed point (DERIVED, the advisor's study; its closing line waits on the owner's word). The fall of a free packet at rest in a content gradient g depends on its pair's gap $\epsilon = (\text{den} - \text{num})/\text{den}$: $a/g = (1 - \epsilon)/(3(1 - \epsilon/2))$, exact from the band, 0.2667 at [4000, 6000] and $1/3$ for a light pair, so the universality of free fall is one family's, and two families with different gaps would fall differently by up to a quarter where nature's materials fall alike to 10^{-13} ; the inertia of a family is $3 \text{ den} \sin \omega_0 / \text{num}$. And a body is never smaller than its quanta's

Compton length, $0.577/\omega_0$ Links, since a record localised to a size R carries the band's kinetic shift about $1/(6\omega_0 R^2)$ against a well of at most $\omega_0/2$ under the horizon; so the law's proton, a cloud of the electron's family, is at least the electron's Compton length across (0.69 Links at $\cos\omega_0 = 2/3$, the cloud's radius $2\Gamma/C$ beyond it), where nature's proton is 460 times smaller. A heavy family with a Compton length of one Link gives a compact proton and breaks the fall's universality by the quarter; a light heavy family gives the fall and not the size; the two constraints together point to a proton of about three heavy quanta at a Γ near 18, which fails the fall: within Rule3 with the well as the binding there is no universe with nature's sizes and nature's fall, and the road closes. Under the rule that clicks are compared with clicks, the proton's radius and the electron's Compton length are both board numbers, so the statement is written only from a scattering look, short matter waves off a cloud onto a screen, whose blind row is that the cloud's pattern is as wide as the electron's Compton pattern or wider; the study stands as algebra until that look's rises are in. The study's record: the compact branch exists for a middling pair [5000, 6000] too, tighter and with its fall more suppressed ($2\cos\omega_b = 1.7159$ at 2,000 at the Node, 2,492 in all, the fall 0.101) [1, the rows against nature, (v) (2); what is open, items 1 and 3].

The lightest body, and the pair tied to Γ . Under the two rows a cloud stands only within the range R : under [600, 601], $R = 10$, the cloud of 4,000 stands ($2\cos\omega_b = 1.3367$, the rms radius 10.4, its well 0.995 and its inertia 1.02 of its count, its fall 0.975) while the clouds of 1,836 and 1,000 do not (their Poisson radii 12.2 and 12.7 Links, over the range); the binding holder holds no cloud lighter than about 1.5 times $2\Gamma/R$, so there is a lightest body, the lightest cloud is the lightest baryon, the binding holder's own quanta are the mesons, and the electron is never a body. The cloud of 1,836 is not bound under $R = 10$ nor under $R = 15$ ([1350, 1351], 1.33320, under the top by a hair) and is bound under [2400, 2401], $R = 20$ ($2\cos\omega_b = 1.33394$, the rms radius 11.9, its well 0.997 and its inertia 1.011 of its count, its fall 0.988 of a free packet's), so the file's pair is [2400, 2401] and R is tied to Γ by the proton, one number fewer in the file (COMPUTED) [1, the rows against nature, (w)][3, the body of nature is the cloud].

The charged cloud, and the proton against the neutron. The charge is the booking W , read into the pace at the weight 1 as gravity's well is, so a cloud whose every quantum carried charge would have a hill of the sign holder equal to its gravity well and would not bind; the proton stands as a cloud of 1,836 quanta with a total W of one unit, its imaginary part a thousandth of its real one, one mode of one sense whose rotation is a thousandth elliptical. Derived beside it, a reading of the algebra set beside nature's number, no click and so no finding against nature: the charged cloud is bound a little less than the neutral one of the same count, by its own hill, so it is heavier by about 10^{-5} of its mass, where nature's neutron is the heavier, by 1.4×10^{-3} , opposite in sign and smaller [1, the two branches; the rows against nature, (z)].

5.4 What a stone is, and how a body is computed

In the law's words, from one quantum to a stone, each step a line of this paper and none a declaration (DERIVED where it follows from the lines, ASSUMPTION where the owner's decisions are named; the bodies' looks are parked, so nothing of this has run, Section 7). (i) *A quantum of matter* is a record of the matter pair $[m, \Gamma]$ at rest, rotating at $\cos\omega_0 = m/\Gamma$, one real line, neutral; free, it disperses by its band and carries one quantum in its share; the electron is one such quantum laid with a sense, a plane, and nature's lightest body is not a body at all (Section 5.3). (ii) *A body* is many quanta of one family whose record stands: its form $D_i \text{ div } T$ is written each interval into the binding holder's row at its own Nodes, the row rests at the screened sum with the range R , every Node of the record reads that level into its paces, and in those paces the record's rotation lies above the top of its

family's band, so no part of it can travel away: the clicks cannot leave. A body is the fixed point of that loop, the record whose own well returns it, and it has no law of its own: the one rule written for the Node makes it, as the support of a mode of that rule (Section 5.1). Its count is the share of its record over its Nodes, a reading; its mass in every field it sources is its well, the form over T summed; the wide fixed point, the cloud, is the body of nature, and the compact pixel at the horizon is collapsed matter (Sections 5.2 and 5.3). (iii) *How a body is computed*: by the law's own iteration and no other formula, the generator (Section 5.5). Start from a spread of the declared count over the Nodes; write the wells that spread lays into the held rows and let them rest; find the record's top mode in those paces by Rule3's own read act iterated; scale its two levels so that the share carries the declared count; write the wells that record lays; repeat until the levels repeat, in integers, the fixed point of the division act. The branch is chosen by the start's width alone, a wide start reaching the cloud and a narrow one the compact pixel, and the amplitude unit comes from the integer width, never written. In the engine this is `tools/pixel.mode.py`, which writes the body's levels into the mode file beside the world; the world file declares only the body's family, its Nodes and their counts, and the gate at the start checks the declared count against the share the laid record reads, a reading and no lay [1, the generator; the four lines of the body][3, the generator is Rule3]. (iv) *A nucleus* is bodies within the binding holder's range of one another, $R = 20$ Links at the pair [2400, 2401], bound by the screened sum, the law's strong force, with the holder's own travelling events as its mesons; the proton is a cloud of 1,836 quanta, the count an input of nature (Section 5.3); *an atom* is a charged cloud, a plane of ellipticity $1/1,836$, with an electron held in the sign holder's well it writes, the two signs of the charge waiting on the sign holder's rotation (Section 3.6). (v) *A stone* is many such bodies standing in one another's wells, within tail reach, in one glow of the held rows: who is bound to whom is read from the nesting of the levels at the Nodes and never declared, and the whole is one body for the clicks, since one click at one place tells that the whole functions as one (the owner's reason for the click, Section 1). A stone moves by Rule3 alone, its quanta following its record's current with no accumulator of its position; its clocks are slowed alike by the well it stands in, once for the clock, and it falls as a cloud falls, as a free quantum falls to the order $(C/\Gamma)^2$, Galileo's fall and the equivalence principle for the bodies of nature (Section 6, (h)); at rest and symmetric it clicks nowhere, cold; it is seen because its surface's bodies are instruments in the meeting form, absorbing light at their rotations' resonance and writing light of their own when they breathe or move (Section 3.8); it sources gravity with its well and resists with its quanta, and the three masses of a stone are one to the order $(C/\Gamma)^2$ because its bodies are clouds (Section 5.3). What the law has not shown of a stone, by name: no body has run on the engine under the lines of this paper (the wells parked at the owner's word, Section 3.4); the proton's size stands as a finding against nature (Section 5.3); and the well's three lines, the count as the baryon number, the mass as the well and the mass defect, wait to be restated when the wells return (Section 5.2) [1, one thing, two faces; what a body does, in the words of the clicks; the click joins and parts][3, what a body is; the click].

5.5 The generator

The generator builds every world from zero, declares nothing and holds no float: the fixed point of the binding row by the law's own iteration, the mode under the wells its form lays and the wells from the mode, until the levels repeat; the two levels scaled together so that the weighted share the record lays carries the body's count; the held rows at their rest, the fixed point of the division act with its remainder, computed once before the first interval and never in it; a body laid with a sense carrying its second level pair, the record a quarter period on; a message laid by the file's keys alone, the wave under its envelope with its level one interval earlier, every cosine by Rule3's rotation act.

The branch is selected by the start's width alone: from a wide start the iteration reaches the cloud, from a narrow start the compact pixel, two fixed points at one total, and the iteration goes to the one its start is nearer to, never across. The amplitude unit A is derived from the integer width as the largest amplitude at which Rule3's total stays inside it, never written [1, the generator][3, the generator is Rule3].

6 The forms of nature the line arrives at

Each form below is arrived at from the line before any run and is a prediction of the law, never a result; the form of nature beside it is on the comparison side and is never an input. The coefficients of Eq. (1) were written for the Node: a held level lowers the clock's pace once and the Link's pace once more, with no metric, no boost and no potential in them; what follows is what those coefficients do to a packet, read by the geometric optics of its band. $U_b = c_b/\Gamma$ is the potential at the point named, the clock's shift per level [1, the rows against nature].

- (a) *The redshift* (DERIVED). Two clocks of one family at the levels c_1 and c_2 have their tick intervals in the ratio of the root of $(1 - 2U_1 + 2U_1^2)/(1 - 2U_2 + 2U_2^2)$, $1 - (U_1 - U_2)$ to first order, for a band whose rotation follows the clock's pace; the band's own slope: at the content x a record of the pair [num, den] at rest rotates at $2 \cos \omega = 2 - 2((\text{den} - \text{num})/\text{den})((1 - x)^2 + x^2)$, so for matter [4000, 6000] $d\omega/\omega = -1.06x$ to first order and the gravitational redshift is the clock's share, $z = 1.06x$; light born by a breathing body in the well leaves with the body's rotation and keeps it in flight (a static content changes no frequency), so a far detector reads the emitter's slow clock. Beside it: Einstein's gravitational redshift [15].
- (b) *The bending* (COMPUTED). At a Node of content x light's index is $n = 1/(1 - 2x)$, $n - 1 = 2x$, the Link's pace reading the content twice (Section 2.5); far from a body its field is the massless row's rest, $L(r) = 3\sigma_{\text{tot}}/(4\pi r)$ beyond a few Links, σ_{tot} the body's wells per interval, so $n - 1 = 3\sigma_{\text{tot}}/(2\pi\Gamma r)$, and a beam past the body at the closest distance b turns, by the geometric optics of the band, by the transverse gradient of the index integrated along the path:

$$\alpha = \int_{-\infty}^{\infty} \frac{\partial n}{\partial b} dl = \frac{3\sigma_{\text{tot}}}{2\pi\Gamma} \int_{-\infty}^{\infty} \frac{b dl}{(b^2 + l^2)^{3/2}} = \frac{3\sigma_{\text{tot}}}{\pi\Gamma b} = 4U_b, \quad U_b = \frac{3\sigma_{\text{tot}}}{4\pi\Gamma b}, \quad (13)$$

twice the clock's share $2U_b$ that a Newtonian pull of the same potential would give, because the Link's pace reads the content twice (DERIVED); the centroid at a receiver L Links beyond is shifted by $4U_b L$ toward the body. The ratio that is nature's: the bending is $2x$ and the redshift $1.06x$, so light is bent by 1.9 times a matter clock's share (exactly 2 against the clock's own share x , the 1.06 the band's slope at [4000, 6000]), against nature's 2, with no chosen number in it. By the geometric optics of the band from the coefficients themselves (the script beside the paper, both forms of the self coefficient): the deflection is 2.02 times Newton's $2U_1/b$ with the clock to the first order or the second, Einstein's $4U_1/b$; with the Link's pace lowered once it would be 1.01 times Newton's. Beside it: Einstein's $4GM/(c^2 b)$.

- (b2) *The magnetic force between parallel charges, Coulomb over γ* (DERIVED, the advisor's derivation at the owner's word, first order in v^2/c^2 , a small gap, long wavelength, uniform motion). Magnetism is no read of the board and needs no line: it is what the clicks show when charges move, from Coulomb's retarded level and the paces' read, the clock once and the Link twice, with Lorentz's factor the band's own. Two like charges moving side by side at the speed v : a moving body reads a level L once in its rest rotation, $\cos \omega_0 = 1 - g(1 - L/\Gamma)^2$ with $g = 1 - \text{num}/\text{den}$, and twice in its kinetic term, $1/m^* \propto (\Gamma - 2L)^2$, so the kinetic term's fall as the level rises reduces Coulomb by $3v^2 = v^2/c^2$, the 3 the lattice's ($c^2 = 1/3$) and not chosen; the moving charge's

level is light's own record at c , whose retarded solution (Liénard and Wiechert's scalar) has a transverse gradient γ times the static one; the product is nature's force exactly,

$$F = F_C \gamma \left(1 - \frac{v^2}{c^2}\right) = \frac{F_C}{\gamma}, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \quad \frac{1}{m^*} \propto (\Gamma - 2L)^2, \quad \cos \omega_0 = 1 - g \left(1 - \frac{L}{\Gamma}\right)^2, \quad (14)$$

Coulomb at rest and Coulomb over γ in the laboratory, the books' magnetic field the factor $(1 - v^2/c^2)$. Derived: the parallel like case, with no magnetic field and no current's read, so the hypothesis of a current's read leaves; not yet derived: the general direction of $\mathbf{v} \times \mathbf{B}$ and induction, the read of a level changing in time, the same ingredients. The blind row for the look of two charged bodies moving in parallel, parked with the pixel: the force over the static one 0.995 at $v = 0.1c$ and 0.866 at $0.5c$. Beside it: the Lorentz force between parallel currents [1, the rows against nature, (b2)][3, magnetism is in the world of clicks].

- (c) *The delay* (DERIVED). A round trip past the body is delayed by $4U_b$ times the path's length within the well's reach. Beside it: Shapiro's delay [17].
- (d) *The perihelion* (COMPUTED). An orbiting emitter at the semi-axis a and the eccentricity e advances $6\pi U_p$ per orbit with U_p the potential at $a(1 - e^2)$: by the same geometric optics the matter pair's periapsis advances $7\pi U_1/(a(1 - e^2))$, 1.17 times Einstein's, with the clock to the first order and $6\pi U_1/(a(1 - e^2))$, 1.00 times Einstein's, with the clock's second order, which is the line (Section 2.5); the bending stays 4 under both. Beside it: Einstein's $6\pi GM/(c^2 a(1 - e^2))$.
- (e) *The moving clock* (DERIVED). A body at the velocity $v = n/W$ (its phase k per Link an exact pair of the file) has its tick lengthened by the dispersion of its own bound band: the tick's factor is $f = \Omega(k)/\omega_0$ with $\Omega(k) = \omega(k) - k\omega'(k)$ the rotation at the moving centre, the phase rate less k times the group velocity, and $\omega(k) = \omega_0 + \Delta(1 - \cos k)$ the body's bound band, ω_0 its rest rotation and Δ its packet's top velocity, both the generator's from the file's counts and pair, no key; to the second order $f = \sqrt{1 - v^2/c_b^2}$ with $c_b^2 = \Delta\omega_0$, Lorentz's factor at the body's own light speed; read as the mean of the front and back click periods $P_v(1 \mp v/u)$ at two resting detectors, u the light's group velocity, $P_{\text{rest}}/P_v = f$. Beside it: Lorentz's factor, at the body's own light speed and not the lattice's, since no boost is among the 48. Lorentz is arrived at as the symmetry of the clicks Outside, not as a symmetry of the lattice [3, the click].
- (f) *The charge* (DERIVED). Like senses a hill, unlike a hollow; a reader of the sense q reads $-q$ times the holder's level in its pace. The coupling goes as the sign: the read by sign enters the pace as q times a level, never as Q/M , so an electron and a proton accelerate alike in one field, and the response in a gradient is the mode's own and not the coupling's. A current's read (the magnetic force, frame dragging) is a hypothesis under its own name and no line of the law. Beside it: Coulomb's force.
- (g) *De Broglie's relation* (DERIVED). A free record of a matter family travels only where its band holds its rotation, $\text{num}/(3 \text{ den}) \leq \cos \omega_b \leq \text{num}/\text{den}$; its wave number is the band's, $\cos \omega_b = (\text{num}/(3 \text{ den}))(2 + \cos k)$, its group velocity $v = (\text{num}/(3 \text{ den})) \sin k / \sin \omega_b$, and to the second order $k = \omega_b v / c_m^2$ with $c_m^2 = \omega_0/(3 \tan \omega_0)$, the family's own light speed. Beside it: $\lambda = h/p$ with the rest rotation as the mass.
- (h) *The fall* (DERIVED). In a pace gradient a free packet or a cloud accelerates at $a = \Delta |d\omega_0/dc| (c_+ - c_-)/D$ toward the well, c_+ and c_- the read level at its two faces D Links apart, so bodies with the same $\Delta \times d\omega_0/dc$ fall alike, and a cloud falls as a free quantum falls to $(C/\Gamma)^2$ (Section 5.3); a compact pixel falls by its tail alone, slower; the universality is one family's, $a/g = (1 - \epsilon)/(3(1 - \epsilon/2))$ with ϵ the pair's gap, and the law has one family of matter. Beside it: Galileo's fall and the equivalence principle, which hold in the law for the bodies of nature and not for collapsed matter.

Table 5 The numbers of one body, the 2,000 pixel at $\Gamma = 6,000$ (COMPUTED from the law's lines, no run): at each distance from the pixel, the field in levels, the clock's share z and the bending of light at that impact parameter, under the binding holder [2400, 2401] and under Poisson's rest; an empty cell is a number the law's document does not give.

Distance from the pixel	The field, in levels		The clock's share z		The bending at that impact parameter	
	[2400, 2401]	Poisson	[2400, 2401]	Poisson	[2400, 2401]	Poisson
1 Link	373	386				
2 Links	189	205				
3 Links	117	133			4.9 deg	5.0 deg
5 Links	62	79	1.09 %	1.4 %	2.8 deg	3.0 deg
10 Links	24	40			1.2 deg	1.5 deg

- (i) *The pull and the orbit* (DERIVED). The massless row rests at Poisson's with the sources and at Laplace's outside a body, $c(r) = 3s/(4\pi r)$ falling to the faces' 0, so the acceleration falls as $1/r^2$ and a circular orbit's v^2 as $1/r$; under the two rows a reader within R reads the binding holder's screened field instead, and a row that reads $1/r$ beyond R takes $E_g = 1$ in its own world. Beside it: Newton's pull and Kepler's fall-off; a flat $v(r)$ would name a missing law.
- (j) *The medium* (DERIVED). Inside a body of pace p light of wave number k is slowed to $v_{\text{in}} = (p/\Gamma)^2 \sin k_{\text{in}} / (3 \sin \omega)$ with $1 - \cos k_{\text{in}} = (\Gamma/p)^2 (1 - \cos k)$, the index v_g/v_{in} , and is reflected whole where $p < \Gamma \sin(k/2)$; through a window moving at w the wave number is matched at the moving face, $\omega - kw = \omega_{\text{in}} - k_{\text{in}}w$, and the speed inside is the band's group velocity at that k_{in} . Beside it: the refractive index and the mirror; Fresnel's drag $w(1 - v_{\text{in}}^2/v_g^2)$ is nature's number beside it and not derived from it.
- (k) *The round trip and Sagnac* (DERIVED). A radiating body receding at v from a mirror at rest reads its returned clicks every $P(u+v)/(u-v)$ intervals for a write every P , u the group velocity at the wave number that solves $\Omega(k) + kv = \omega_b$; a radiating body moving at v along a ring of N Links reads its two arms $2Nv/(u^2 - v^2)$ apart, $4A\omega/u^2$ for the ring's area A and angular speed ω to first order. Beside them: the two-way Doppler ratio and Sagnac's $4A\omega/c^2$, with u in place of c .
- (l) *A free record keeps its wave number* (THEOREM). Rule3 is translation-invariant, so a free record keeps its wave number and rotation exactly and a radiating body's period is read unchanged at every distance, the record's rows widening by the dispersion while its count does not move, the rounding's walk at most one unit of level per Node per interval. Beside it: Hubble's drift, which the law does not give from a fixed Γ ; the law's own line for an expansion is the hypothesis *Gamma is not constant* (Section 8.4).

The numbers of one body. For the 2,000 pixel at $\Gamma = 6,000$ (COMPUTED, the advisor's derivation): under Poisson's rest, the bending $0.26/b$ radians, 3.0 degrees at $b = 5$ Links and 1.5 at 10, a $\lambda = 8$ packet passing at $b = 6$ turning by 0.043 radians, a transverse shift of 1.3 Links over 30 Links of path; a matter clock five Links from the pixel ($x = 0.013$) running 1.4 percent slow against a far one. Under the two rows the massless row's field is over E_g and a reader within R reads the binding holder's screened field instead; Table 5 sets the two side by side [1, the rows against nature, (a) and (b)].

The bending stays 1.9 times the clock's share under both; the lattice computation of it to three digits, a pixel of 2,000 and a packet passing at five Links, the packet's deflection against the clock's share at the same Node, is the first proof-grade number the law offers (Section 8): 2.00 within the rounding is the direction confirmed, 1.9 sharp a lattice correction to state [1, the rows against nature, (b) and (v)].

How the forms are reached, not assumed. Nothing above was put into the coefficients: no metric, no Lorentz transformation, no potential and no force. The one thing the law says is that a held level slows a Node's clock once and its Links twice, and that a record's rotation follows its band. The bending's 4 and not 2 is the Link's second factor; the perihelion's 6π and not 7π is the clock's second order, the composition of the slowing unit by unit; Lorentz's factor is the Legendre transform of a bound band read at its moving centre. Where the law's number differs from nature's form at the declared numbers, the difference is the law's own, read blind: the bending's 1.9 against 2 (the band's slope at the matter pair), Sagnac's u against c , Fresnel's drag not derived.

7 The two experiments, derived and not run

The paper carries no experiment. The owner's word of 2026-10-01, verbatim in translation: "until we have experiments that worked and were proven, meaning that both the advisor and the reviewer confirmed, the paper is without experiments at all; it is only about the algebra, the algebraic derivations, all the ways we work here, which formulas we reached by derivation and which we did not" (ASSUMPTION, the decision) [3, the paper]. So no run's number stands in this paper: no count through a screen, no S of any run, no arrival interval read, and no figure of a measured row. What stands is the two experiments the paper names, two and no more, each as the engine's world, the implementation, and as the law's own derivation of what it must give, the advisor's by all the rules, marked DERIVED or COMPUTED and never a result; their numbers enter the paper when their runs have worked and the advisor and the reviewer have confirmed them. Nothing in an experiment brings the right result by construction, and a blind file is never touched after its run [3, nothing in an experiment brings the right result by construction].

7.1 The two slits: the world

The bright world alone. The emitter is a laid message, the emitter's events, a packet of the light family (the holder of the sign's own real record, one real line), no body: a wavelength of 8 Links along $+x$, $k = \pi/4$, $\cos\omega = 0.9024$, the period 14.1 intervals, the group velocity 0.547 Links per interval at that wave number, $(1 - 0.083 k^2)/\sqrt{3}$ (light's speed at long wavelength is $1/\sqrt{3}$); its levels laid at the first act as $\text{now}_i = b e_i \cos(kx_i)$ and $\text{before}_i = b e_i \cos(kx_i + \omega)$, the record one interval earlier, with $b = 1,328$, about ten quanta per Node of the plane record, under a raised cosine along x over $x = 4$ to 16 centred at $x = 10$, flat across $y = 4$ to 44 with edges of two Nodes; a real packet, neutral. The wall is not matter but the board's own face rule declared inside the board: the inner face at $x = 20$ with two gaps of three Nodes, $y = 17$ to 19 and 29 to 31, the slits' centres $d = 12$ Links apart, a declaration of the file like the wraps and the detectors' regions, no number in the engine. The board is flat, $45 \times 48 \times 1$, the third axis folded, its y faces open and its far x face receding, so that the screen, the board's last column $x = 44$, $L = 24$ Links from the wall, is backed by the receding face itself and is opaque. The screen is twelve declared detectors, the regions of four rows of that column from the row 0, the smallest region the law allows at $\lambda = 8$, each reporting what it sees; the draw's seed, 24, is declared in the blind file. The run is 130 intervals, the screen read in the window of intervals 45 to 120. The universe file is the light-only one, the rule's own with the three holders' divisors at 10^6 , so that light's own wells round to nothing (Section 8.5, (1b)); in the engine the world is one file written by a builder from one design file that holds every number above, with its mode file and its blind file beside it, `examples/events/two_slits/` [1, the rows against nature, (g)][3, the two slits]. The folded board is the experiment exactly and no special case: for a record uniform along z the folded axis returns the Node itself through both z Ports, so

light's band on the flat board, $2 \cos \omega = (2/3)(\cos k_x + \cos k_y + 1)$, is the three-dimensional band at $k_z = 0$ (DERIVED) [1, the rows against nature, (g)].

7.2 The two slits: what the law says it must give

Three readings of the one world, each a derivation written before any run and never touched after it. (a) *The fringe.* The far field's spacing is $\lambda L/d = 8 \times 24/12 = 16$ Links; at $L = 2d$ the fringe is the near field's, the two gaps' path difference reaching half a wavelength 8.7 Links from the centre and a whole one 22 Links out, so the minima fall at the rows 15 and 33 and the outer maxima spread over the rows 0 to 7 and 40 to 47 (DERIVED). The row reached from the families' mathematics alone: Rule3 for light, num = den, is $a_{\text{next}} = (\text{the six Ports' sum})/3 - a_{\text{before}}$ at the vacuum's paces, and stepped in real arithmetic on the very world it gives 280.8 quanta through the screen over the window, per region from the row 0: 20.5, 19.7, 27.1, 16.2, 9.4, 41.1, 47.7, 14.6, 12.2, 27.1, 21.6, 23.4 (COMPUTED, the advisor's); the Huygens sum of the two gaps with the obliquity factor gives 273, three percent under, the same central fringe and the lattice's own wings (the regions 3 and 8 at 16.2 and 12.2 against the continuum's 22.6 and 19.2), a lattice number in the same world. The engine's integers must agree with the real line to the rounding, about a third of a quantum per region; a larger difference is a bug of the engine and never a finding. (b) *The clicks.* The screen's shares rounded to whole quanta by the largest remainders are the expectation; the row drawn by the shares at the declared seed is the clicks, with the draw's scatter $\sqrt{Np(1-p)}$ per region; each row is named for what it is and never for the other. The success criterion, written here: the central maximum against its two minima at 5.7 standard deviations of the draw's scatter, so that the fringe is visible beyond the draw by any seed, and the total within a few percent of the real line's. (c) *The wager, the lattice's own number.* The interval of the clicks' peak at the screen: the law's own line stepped in real arithmetic on the world gives the screen's inflow peak at the interval 68 (the centroid 73) on the lattice, against 63 (the centroid 70) for a continuum wave in the same geometry, the same world scaled toward the long-wavelength limit; the click line resolves the five intervals (COMPUTED, the advisor's). Under the meeting's click nothing moves here: light is one real line with nothing to pair, the share is the record's own inflow, and the one form of Section 3.2 has the two slits as its case with nothing to pair [1, the rows against nature, (g)][3, the click is the meeting of the future with the past].

Two worlds the paper's drafts carried leave: a world with one gap closed by the wall's own face, which is a closed slit by construction and no which-way measurement (no detector credits the gap; the path marked without absorption is beyond the instrument the law has, and the paper says so), and a dilute world whose blind row was rescaled after its run, which no blind file may be from here on [3, nothing in an experiment brings the right result by construction].

7.3 Bell: the world

The meeting world alone. Entanglement in the law's words is one event with two trees: the pair is laid as the pair family, two real lines of light never summed at a Node, nature's HH and VV, with equal parts, from one symmetric source, one beam to each side; each side's detector is a declared analyser, a region of two ports at least half a wavelength across with a declared basis given as a pair of integers (p, q) , the $+$ port reading the parts as $pU + qL$ and the $-$ port as $-qU + pL$ at every one of its Nodes before the density, exactly orthogonal, so the single-side marginal is one half to the bit for any pair; the CHSH settings as the pairs (1, 0), (1, 1), (12, 5) and (5, 12), 0, 45, 22.62 and 67.38 degrees, no cosine and no root in the engine. The credit is the one form of the click

(Section 3.2): the joint share pairs the parts by their label across the two sides through the root,

$$J(p_A, p_B) = \left| \sum_k c_k e_k(p_A) \alpha_k(p_A) e_k(p_B) \beta_k(p_B) \right|^2, \quad c = (1, 1), \quad (15)$$

α_k and β_k the part k 's amplitude at the port on each side, $e_k(p)$ the port's coefficient on the part k , an integer whose square is the joint density, the four densities over their sum giving E as a rational with the norms cancelling; never the square of a local sum, which one detector could do alone and which gives $E = \sin 2a \sin 2b$ and S at most 2; one draw per pair ordered in time, the first side from its marginal and the partner from the conditional. A comb with two gaps and a port shift, the paper's earlier instrument, is not Bell: a port of half a wavelength spans about 80 degrees of the comb's phase and cannot read it, the uncertainty relation of Section 3.3 itself, so the comb and the two gaps stay the two slits' world's (ASSUMPTION, the owner's words of 2026-10-01; the meeting round of the engine, gated by one dedicated test of the reader: the pairing through the root against the local sum failing by name, the determinism of equal parts, the two local numbers as the lower fence) [3, the detector declares its basis angle; the generic Node is closed for building].

7.4 Bell: what the law says it must give

Three layers, two of them the board's algebra and the third the law's word. (1) Rule3 gives each part's amplitudes at the ports: the two parts of one beam, laid equal, are two records of one family stepped by the same line along the same path, bit-identical at every Node at every interval by determinism; a difference between them is a bug of the board or an asymmetry of the geometry, named by the diagnostic of the parts' relative amplitude and phase at each detector (THEOREM). (2) The meeting with the linearity of Rule3 gives the joint amplitude: the realised click at A's port is the past, a weighting of the parts by A's analyser, read back to the root where the two parts were one event and forward to B where the future is B's analyser's weighting; the meeting at B multiplies the past and the future part by part and the root sums the two branches, $\sum_k c_k e_k(a) e_k(b) = \cos(a - b)/\sqrt{2}$ for equal parts (DERIVED). (3) That the share is the density of the meeting, the square, is the owner's declaration, Born's rule for a pair: nothing in Rule3 forces it, and the clicks force it, since it is the least non-local credit that reaches nature's number with every quantum counted (ASSUMPTION). From the three, with the pair's common phase θ uniform over the pairs, A's + port at a and B's at b ,

$$\begin{aligned} \text{the meeting:} & \quad P(++) = \frac{1}{2} \cos^2(a - b), \quad E(a, b) = \cos 2(a - b), \quad S = 2\sqrt{2}; \\ \text{a local credit by the shares:} & \quad P(++) = \frac{1}{4} + \frac{1}{8} \cos 2(a - b), \quad E = \frac{1}{2} \cos 2(a - b), \quad S = \sqrt{2}; \\ \text{a local credit by the sign:} & \quad E = 1 - 4|a - b|/\pi \quad (|a - b| \leq \pi/2), \quad S = 2; \\ \text{the credit by the contrast:} & \quad E = \cos 2(a - b) \text{ at the efficiency } 2/\pi, \quad S = 2\sqrt{2}, \end{aligned} \quad (16)$$

the second by averaging $\cos^2(a - \theta) \cos^2(b - \theta)$ over θ , the third by the band of width $|a - b|$ in each half-period where the two signs differ, the fourth by the integral of $\cos 2(b - \theta)$ over the half-period where $\cos 2(a - \theta) > 0$ against the mean weight $(1/\pi) \int |\cos 2u| du$ (DERIVED, the advisor's four derivations). So $E(a, b) = \cos 2(a - b)$ under the meeting, and at the integer pairs $E = 0.7042, -0.7042, 0.7101, 0.7101$ at $(a, b), (a, b'), (a', b), (a', b')$, $S = 2.8286$ to the rounding of the ports' integer sums, every pair credited on both sides, the efficiency 1 and the loophole closed by construction, the marginal one half on each side whatever the partner's pair (COMPUTED). What the board cannot give without the declaration, the second and third lines of Eq. (16), two theorems of the local credits on the same world (THEOREM): a credit of each side by the shares of its ports

gives $S = 1.41$, a local draw never reaching Bell's bound; a credit of each side by the sign of its larger port, the pair's phase a shared classical variable, gives the sawtooth, whose CHSH sum is exactly 2 for every ordered set of four settings within one period, the bound the sign's own. The loophole's form, the paper's earlier Bell, superseded: a two-port detector that credits by its contrast and counts nothing where its two ports' shares are nearly equal gives $E = \cos 2(a - b)$ exactly for every visibility V at the efficiency $(2/\pi)V$, the uncounted pairs those near the contrast's zero, the detection loophole of the Bell tests, a local shared-variable form that reaches nature's cosine only below the loophole's bound (DERIVED); it reported a formula and is not run as Bell. Bell carries no wager against a continuum: its number sits in the credit's rule by the owner's declaration, with the two local theorems as what the board cannot give without it, and the paper says exactly that. Nature beside the derivation: $E = \cos 2(a - b)$, and $S = 2.42 \pm 0.20$ in the loophole-free test of 2015 that measured the sum [9]; the two photon tests of that year violated the Clauser-Horne-Eberhard form of the inequality at detection efficiencies near 0.75 [18][19]. No signal passes between the sides: the detectors read locally, the root joins by linearity, and each side's marginal is one half; no joint credit by an instrument's declaration (two detectors are never at one Node) and no settings chosen by the board's own past (no mechanism); the superdeterministic world, the settings set by bodies on the board that share the source's history, is a world under its own name and not before this one, two detectors correlated before the click lying inside Bell's λ and raising no bound [1, the rows against nature, (h)][3, the click is the meeting of the future with the past; the detector declares its basis angle].

8 The constants, and what is open

8.1 The constants

The causal bound is one Link per interval, and light's own speed at long wavelength is $1/\sqrt{3}$ Links per interval (Section 2.2). The quantum of action is one click, its energy the quantum's norm T of its family. The quantum of distance is the Link, of time the interval; nothing between two Nodes or two intervals is observed. Newton's constant is not declared: a body whose wells are s quanta per interval makes the massless row's rest $3G(r)s$ at r Links over the row's divisor, $0.7582s$ at the Node and $3s/(4\pi r)$ far away, and the level slows the clock by the potential $\Phi = -c/\Gamma$, so (DERIVED)

$$G = \frac{3}{4\pi\Gamma E_g} \text{ per quantum of well, in Links and intervals,} \quad (17)$$

a reading of Γ , the file's E_g and the lattice, and no number of the law; a world at a smaller Γ has stronger gravity per unit of content, and the ratios between rows carry G once and cancel it. The row's source moves by whole quanta of well and never by less, and the field itself is no click. The vacuum content, the massless row's declared rest, is the law's cosmological constant, 0 by nature's clicks (Section 4.6) [1, the constants].

The free numbers. Under every divisor 1, G and α are no numbers of the law but readings of Γ and the GameBoard's geometry; under the two rows the file holds gravity's E_g beside Γ , T and the matter pair's m , the binding holder's gapped pair (its range R tied to Γ by the proton) and the vacuum content, 0 by the declaration. Γ is free and large against a proton's count, as nature's Link is small against a proton, and no ratio fixes it yet (item 19 below); α stays a reading; m is the one integer of nature beside them, bounded by the band and the reach [1, no free number but the mass].

T is physical through $\sqrt{T}/\Gamma$. T is not the resolution only, though the rounding's floor stays (2^{19} for a hundredth of a quantum per unit at the pixel): the level of one quantum of light is $\sqrt{T}/\sin\omega$,

from $b^2 \sin^2 \omega / T = 1$, and the pace reads levels, so light's action per quantum on a body's content is $\sqrt{T}/(\Gamma \sin \omega_L)$, 0.07 of Γ for one quantum per Node at $\lambda = 8$ at $T = 2^{15}$ and $\Gamma = 6,000$ (0.03 the floor at the shortest waves, 0.28 at $T = 2^{19}$); matter's well per quantum is 0.76 levels, so at these T and Γ light acts on a pace 550 times more per quantum than matter's gravity. The pair (T, Γ) holds one physical number, $\sqrt{T}/\Gamma$, light's coupling per quantum, and nature's is small; what fixes it is open beside Γ [1, what is open, item 9].

8.2 The constants the law derives with no free number

A constant of nature is derived as a dimensionless number, or as a dimensional one once the units are fixed from nature; five candidates, in the order of their computation, each written before its run and each written as it comes, against nature or not [1, the rows against nature, (v)]:

- (1) *The bending factor of light*, the bending over the clock's share, Einstein's 2 against Newton's 1: 1.9 on the lattice in the first approximation (Section 6, (b)); the lattice computation to three digits is the first proof-grade number, 2.00 within the rounding the direction confirmed (COMPUTED in the first approximation).
- (2) *The strength of gravity against the quantum*, $Gm^2/(\hbar c)$: under every divisor 1 the coupling of one quantum to another through the content holder is of order one, against nature's 1.75×10^{-45} for the electron and 5.9×10^{-39} for the proton, and the same number read between two charged clouds is the hierarchy (Section 4.5); decided as the two rows with the caveat, the hierarchy the file's one number, E_g about 3×10^{42} for nature, and the charged cloud look reads it as $E_g/1,836^2$; the reading of scale beside it, structural within Rule3 with the well as the binding by the second family's study (Section 5.3), written only from the scattering look (COMPUTED under every divisor 1; ASSUMPTION under the two rows).
- (3) *The fine-structure constant*, α : the sign holder's well of one quantum with the sense against the content holder's well of one quantum, both at the divisor 1, the electric against the gravitational pull between two electrons, nature's 4.2×10^{42} , and α itself as light's coupling to the charge, to compute, with its distance from $1/137.036$ (open).
- (4) *The mass ratio* $m_p/m_e = 1,836$ is the count's ratio of a cloud to one quantum, an input and not derived (Section 5.3), and fixes no Γ , so (1), (2) and (3) are computed as functions of Γ until the line that fixes it is written.
- (5) *The lattice constants*, $3G(0) = 0.7582$, $3/(4\pi)$ far away and light's $1/\sqrt{3}$ Links per interval, the six Ports' own, exact and not nature's, entering every derivation above (COMPUTED).

8.3 The hypotheses under their own names

Every hypothesis stands outside the law under its own name until it passes the three tests or the owner admits it; each says what would decide it [1, the hypotheses under their own names].

Hypothesis 1 (the conversion, the weak force). A click that takes a quantum of one family and gives it as quanta of others, the rotations adding, is the whole weak interaction, one Node's act, no field and no massive vector, and no family's row declares anything for it.

Hypothesis 2 (Gamma is not constant). $\Gamma_{t+1} = \Gamma_t + 6$, the step the six Ports of a Node and no free number; the rule's physics depends on the ratios c/Γ and m/Γ alone, so every pair steps with Γ and the wall and the amplitude unit are derived from Γ_t each interval; a pixel of a fixed count falls under the rising edge and dissolves, the distinct bound counts grow with Γ , light does not drift while the clocks of pixels do, the rule's Hubble rate $H = (6/\Gamma) |d \ln \omega_b / d \ln (c/\Gamma)|$, and $\Gamma = 6t$ reads the universe's age in intervals. The time road's price: a stepping Γ is Dirac's road, G falling

as $1/\text{age}$; nature bounds G 's drift today under 10^{-13} per year against 7×10^{-11} per year for a G proportional to $1/\text{age}$, and in the law a coupling that decays weakens the bodies' binding with it unless the binding is a row of its own, which the two rows made it [1, Gamma is not constant].

Hypothesis 3 (the three distances are the rule's). That α , m_e/m_P and m_p/m_e are derived; the algebra is closed on one root, two named assumptions, one assumption of the universe and one hypothesis, with no number but gravity's E_g , and is not closed toward nature by name [1, the tree].

Hypothesis 4 (the hill holds the body). A body stands as a stable extended fixed point where a held row read with the opposite sign, of reach shorter than the hollow's, is held beside the hollow: below the hill's reach both are Coulomb-like and the hill forbids the collapse, between the two reaches the hollow binds, and the body's size lies between the reaches. Decided by a run of the generator with the third row.

Hypothesis 5 (the cooling). A body above its minimum sheds its excess as quanta of its own family, messages of matter, and radiates light by its write while it breathes, the shedding stopping at the stable body theorem's minimum for its count; the engine sheds by the count's line alone, and whether a body settles at the minimum is a run's finding. The number of quanta a cloud gives to settle: for a Gaussian cloud twice too wide in its own massless well, 13 percent leave on a chain, 20 on a plane, 23 in a box (DERIVED, a blind expectation) [1, the number of quanta a cloud gives to settle].

Hypothesis 6 (the anti-body). The anti-body was the same record with its count laid as -1 at its Nodes, a body of holes moved by the count's line; that lay left with the lay, the count being the record's share, which has no sign to reverse, so the anti-body has no line today and the antineutron row waits for a line of the record, distinct from the neutron.

Hypothesis 7 (the second level's amplitude is a count of its own). The second level's amplitude quantised as a count of its own, a circular quantum the unit and a body's charge its number of circular quanta, so that $|q_p| = |q_e|$ would be derived where the law as written leaves a body's charge continuous (Section 3.6); no line of the law as written does it and Theorem 2 may not allow it.

Beside them, by name: the third way of gravity (the massless row's level as a count of resting clicks relaxed each interval, which trades the exact inverse or the causality; Le Sage's shadow refused, the drag on a moving body and the absorbers' growth), a controlled phase of order one between two records (which Theorem 2 excludes from the acts alone), and the sign holder rotating the two-part record, the declared act of the file beside the plain read, under which like and unlike senses differ with no sign read from a Node's state, a round of its own after Bell's first look (Section 3.6) [1, the fork, decided; the sign holder rotates the two-part record].

8.4 What is open, and what is not claimed

The law's document keeps its open items by name; an item the law's own words closed stays as one line saying why, a decided line stands against nature until tested, and a line decided with a reservation stands until one of its reopeners [1, what is open].

- 1 to 3. *The proton and the electron, the two branches, the fall and Q/M* , closed: the body of nature is the cloud, its three masses one, the proton a cloud of 1,836 and the electron one quantum, m_p/m_e the count's ratio fixing no Γ ; the proton's size stands beside it, a body never smaller than its quanta's Compton length where nature's proton is 460 times smaller, written from the scattering look; the coupling through the pace is universal and the response is the mode's own, so the equivalence principle holds for the bodies of nature and not for collapsed matter, whose fall is the fall look's blind row at detectors; the universality is one family's (Sections 5.3 and 6).

4. *The lone quantum's detection*, decided: the click ends nothing, and the detector is a declared instrument whose credit is the instrument's, so exclusivity and Born's rule are a declaration and no finding of the law, the two slits' dilute world rising click by click by the declaration (Section 3.9); the count is the record's share (Section 3.4), and the algebra of the exclusivity stands in the documents under its own name.
- 5 to 8. *The colours and the third*, closed as absent; *the universe expands*, closed as a statement about boards (a board closed on every axis has no rest and its level drifts, the finite board's word); *the unread parts*, closed as out; *the builds*, closed as decisions.
9. Γ and T : Γ the one scale of the law, as the Planck length is nature's, free and large against a proton's count; T physical through $\sqrt{T}/\Gamma$; what fixes either is open.
- 10 to 12. *Radiation reaction*, closed by the sense's build (whether a breathing body cools is a run's finding); *the mass defect*, closed as the law's reading and parked with the pixel (Section 5.2); *the neutron and the antineutron*, reopened with the lay (Hypothesis 6).
- 13, 14. *The generator's branch and the start's sources*: the start sourced by the wells, the fixed point of the wells under the rest they make, is the pixel round's line; *the tensions' start*: the start sources the time parts alone, so the tensions start at 0 and ring up to their rest over the first intervals, a transient that is no wave of the body.
15. *A cross-axis read or a current's read*, left, derived: magnetism needs no read of the board, the magnetic force between parallel charges being Coulomb over γ from the law's own lines (Section 6, (b2)).
- 16 to 18. Whether a body at rest jitters when its current's swing reaches $T/2$ (on sixteen Nodes the swing over 400 intervals is one percent of $T/2$); the self-source's cubic term, not in the engine; the stable body theorem's Hessian beyond the dilation for a set of held rows and the functional beyond first order, and whether the hill holds the body and the cooling brings a body to it, a run's findings.
19. *What fixes Γ* : the mass ratio of clouds is a count's ratio and fixes no Γ ; the constants of Section 8.2 are computed as functions of it until the line that fixes it is written; that line is open.
20. *The hierarchy of gravity and electricity*, decided with a reservation (Section 4.5): the finding of every divisor 1 kept as the law's word on the road not taken; the reopens the three tests, the bodies not standing under the screened sum, the charged cloud look reading nature's other way, and the finding of scale.
21. *The laid row's inverse*, decided: a held row laid each interval from the wells has no whole-run inverse (the paces of an interval read a level that is a function of a level no later state holds), so every holder of the file steps by Rule3 at its pair and inverts by its own direction, and the whole run goes back in time (Section 2.9); the count and the sense, lines whose inverse the state gave, are readings of the record since the owner's word of 2026-09-30, and the one read the state after the interval did not hold, the reader's sign, leaves with the owner's word of 2026-10-01 (Section 3.6).
22. *The vacuum content*, decided: 0, a declaration fixed by nature's clicks (Section 4.6).
23. *Gravitational waves and the lattice's rotations*, a finding against nature with its look: under the tension's diagonal alone the massless row's travelling events carry the breathing polarisation and the lattice-aligned + polarisation only, and no $\times$, where nature's gravitational waves come in + and $\times$ at every orientation; and the lattice's 48 symmetries stand against nature's rotations at short wavelengths, light's band along an axis against the diagonal anisotropic by about 1 percent at $\lambda = 8$ Links and tens of percent at $\lambda = 4$, falling as $(\text{Link}/\lambda)^2$, a finding where nature's wavelengths are short in Links and otherwise a bound on the Link from nature's isotropy. Settled by the blind anisotropy from the band, the bound on the Link, whether a rotating source radiates on the lattice without the off-diagonal parts and what its events carry against nature's

quadrupole, and a run of a rotating or breathing source on an open board with the massless row read along an axis and a diagonal, its blind row written first.

8.5 Nature's numbers enter the board only as clicks: the calibration series

Every measured number of an experiment (a frequency, a wavelength, a deflection angle, a redshift, a coincidence rate) is first written as a click pattern on a declared detector against a declared clock and only then set beside the board's click pattern; the families' coefficients (the pairs) and the units (Γ and T , what in nature is one Link and one interval, the one calibration the law does not derive) are fixed by that match and by nothing read off the board; the two ratios the law fixes without it are light's speed $1/\sqrt{3}$ Links per interval and the causal bound 1 (ASSUMPTION, the owner's word of 2026-09-30) [1, the reading, nature's numbers enter the board only as clicks][3, the two slits as run are the click experiment; the experiments needed]. What the calibration needs, each a declaration of the file or of the reading and none a number of the engine: the units; the families (which of nature's clicks belong to which row, a click reporting its family, its region and its Ports); the emitter (nature's source as a declared lay of events, or a body's write); the detector (the declared region, its credit rule, what it reports and what it does not; nature's detector efficiency and dead time the same kind of declaration); the geometry in Links, the receding face for an open laboratory; and the reading (the histogram per region, the comb, the coincidence window, applied to nature's data the same way). The advisor's reading per quantity, marked as his: a clock of nature is a click rate, a length a click's place on a screen, an energy or a mass a count of quanta, and a family's pair sets its rest rotation $\cos \omega_0 = \text{num}/\text{den}$, so one click rate of nature per family fixes its pair once Γ and T fix the unit. The series, one number of the law against one click pattern of nature per line, light-only meaning the engine as it is:

- (1) *The two speeds*: the massless row's own events step at the pace 1 at the speed $1/\sqrt{3}$ and light reads the vacuum content $x = c_{\text{vac}}/\Gamma$ into its pace at $(1 - 2x)/\sqrt{3}$, so over L Links the two arrive apart by $2xL\sqrt{3}$ intervals at long wavelength, and from the two bands exactly, $\cos \omega = 1 - (1 - \cos k)/3$ and $1 - (1 - 2x)^2(1 - \cos k)/3$, by 17.9 intervals over 458 Links at $k = \pi/4$ and $x = 0.01$ (COMPUTED); the look is three worlds of one design, light at a content, light at no content and the massless row's kick alone, their arrivals read at a far region as a diagnostic beside light's clicks there, designed and not a result of this paper. Nature's clicks: GW170817, gravity's lead over light at most 7×10^{-16} of its speed [11], which fix $c_{\text{vac}} = 0$ (Section 4.6). Beside it, (1b) *gravity's divisor bounded from below by light's own fringes*: a light packet writes its own form into gravity's row over E_g , and where that well reaches one level the packet's paces vary across it and its fringes move with the intensity, which nature's fringes never do at any laser's power; so E_g is at least the packet's summed form over T for the strongest light ever measured, the light-only worlds of this paper run at 10^6 at $\Gamma = 6,000$ for that reason, and only gravity's divisor is tested by clicks today; nature's number enters as a click too, the bending by a body, 1.75 arcseconds, when the bodies return (DERIVED).
- (2) *Light's dispersion on the lattice*: along an axis $\cos \omega = (2 + \cos k)/3$, the group speed falling as $1 - 0.083 k^2$ (5 percent at $\lambda = 8$, 1.3 at 16, 0.08 at 64); the look two packets of different wavelengths laid together and two detectors, the arrival difference. Nature's clicks: photons of different energies from one gamma-ray burst arriving together, $|\delta v/c|$ below 10^{-16} at 1 TeV, which bound the Link from above, $k = 2\pi \text{Link}/\lambda_{\text{TeV}}$ below 3.5×10^{-8} per Link, so the Link below about 7×10^{-27} metres, the first bound on Γ 's unit, the interval following from the Link and c (COMPUTED).
- (3) *Matter's two slits, de Broglie*, a matter message and no body, when the emitter lays a matter packet at the pair $[m, \Gamma]$: the band's curvature near its bottom, $\omega \approx \omega_0 + k^2/(2m^*)$, gives

$v = k/m^*$; the look reads the spacing $\lambda L/d$ and the packet's speed by two detectors' time of flight, the two clicks giving m^* in Links and intervals, and nature's $h/(mv)$ with the units of (1) and (2) gives m , the matter pair.

- (4) *The light clock*, light-only, now: a packet between two mirror faces, the period $2L\sqrt{3}$ intervals as clicks at one face's detector, a clock with no body; in a declared static well of the massless row its rate falls by x and a screen beside it reads the bending $2x$, the Einstein rows of Section 6 as clicks with the ratio 2 blind; E_g enters only when the well is sourced by a body and waits with the pixel.
- (5) *The binding pair and E_g through bodies*, parked with the pixel.
- (6) *The vacuum content by light's speed alone fixes nothing*: nature defines the metre by c , and one speed calibrates no unit; c_{vac} is fixed by (1).

The order: (1), (2) and (4) now, (3) when the emitter lays a matter packet, (5) parked [1, the reading, nature's numbers enter the board only as clicks].

What is set beside nature, in one place, by kind. Clicks are compared with clicks, and nothing else is a finding against nature; the index the documents keep is read by that rule [3, only clicks against clicks]. *Click rows read*: none; the paper carries no experiment until one has worked and the advisor and the reviewer have confirmed it (Section 7). *Derivations written for the two experiments, to be read against the runs*: the two slits' fringe row from Rule3 in real arithmetic, 280.8 quanta over twelve regions at the near field's places, and the lattice's arrival interval, 68 against a continuum wave's 63 (Section 7.2); Bell's $E = \cos 2(a - b)$ and $S = 2.83$ under the one form of the click with every quantum counted, beside the two local theorems 1.41 and 2.00 and nature's 2.42 ± 0.20 measured without loopholes (Section 7.4). *A number fixed by nature's clicks*: the vacuum content, 0, by GW170817 through the derivation of the two speeds from the two bands (Section 4.6); gravity's divisor bounded below by nature's fringes not moving with the intensity, and the Link bounded below 7×10^{-27} metres by the dispersion of a burst's photons, bounds and no numbers yet (Section 8.5). *The documents' looks with their click rows, not experiments of this paper*, each a claim of the law beside nature's click pattern, written only when its rises are in and confirmed: the bending of a packet past a body against the clock's share at the same Node, 2.00 within the rounding or 1.9 sharp, nature's 1.75 arcseconds the click that will fix E_g ; the light clock's rate in a declared well with the bending beside it; the magnetic force between parallel charges, 0.995 at $0.1c$ and 0.866 at $0.5c$; the two signs of the charge under the sign holder's rotation, like repelled and unlike attracted at one size; gravitational waves' polarisations and the lattice's rotations (item 23); the charge's unit, the charges of laid bodies read by their deflections; the proton against the neutron and the hierarchy, the charged cloud look; the antineutron, which waits for a line of the record; the rotation curves and the redshifts; the proton's size, the scattering look; the compact body's fall, the fall look's rises at detectors; the resonant absorber's one click, the absorption look, the first look of a body as an instrument. *Left as readings, never compared with nature*: light's speed against the causal bound (the bound is no click of nature's); the three masses along the two branches; the charged cloud against the neutral one until its look; the share's drift under a moving well.

What advancing the paper needs. Everything nature measures is clicks, and every number of the law is a number of a file: the Node clock Γ , the quantum's action T , the matter pair's m , the binding holder's pair, gravity's divisor E_g , the vacuum content, and the families themselves, each a row (Section 4.2). Between the two there is no dictionary yet: the paper does not know how a click pattern of nature's is turned into the file's numbers and rows, which family of the law a pattern names and at which parameters, and the rows against nature of Section 6 are read at

the looks' numbers and not at nature's. Advancing the paper therefore needs specific experiments designed for that translation and for nothing else (the owner's word, 2026-09-30): for each number of the file, a click pattern of nature's whose form fixes it, read against the law's blind row at that number, and for each family the click pattern that names it, its clicks' family and axis, its band by de Broglie's row, its holding by the fields it sources. The series above is that list; its first number is fixed, the vacuum content, and two are bounded, gravity's divisor by nature's fringes and the Link by a burst's photons. Until the rest are named, run and read, the law's numbers are the looks' and not nature's, its families are the rule's bands and not nature's particles, and what fixes Γ and $\sqrt{T}/\Gamma$ stays open; the two experiments of this paper are the first two of that series.

Not claimed. Any agreement with nature from a run: no run's number stands in this paper, and the two experiments stand as the law's derivations of what they must give; that nature is so; the Lorentz group, which is no symmetry of the lattice; which meeting becomes the click; that the share is the density of the meeting as anything but the owner's declaration; a controlled phase of order one between two records; the fine-structure constant, the mass ratio and what fixes Γ ; the spin $1/2$; the count of gravitational levels, which is the universe's history; a body's settling, which is the hypothesis of the cooling. Every hypothesis of the paper is named as one.

9 The engine that computes it

The engine, Universe24 [20][2], is the law's implementation and nothing else: a GameBoard of Nodes, every family's NodeState at every Node, and one interval of acts in the law's order, each a call of Rule3 on whole-board arrays in 64-bit integers, every neighbour read through a Port. It holds no number, no formula, no family's name and no flag; every physical value comes from the run's files; no act of the interval takes a square root and no act reads a Node to decide anything. It is six modules under 400 lines each (the Node's acts, the share as a reading, the reports, the receding face's growth, the GameBoard that walks the interval forward and back, and the files with their loader) on a core of Rule3, the Ports and the working bound; every family is a row of the universe file found by its name, and a new family is a row and no code. A Node's state is the generic Node of Section 3.4: per family, per part, two levels and Rule3's remainder, per held part one write remainder, and nothing else, held so by a structural gate in the tests; the count, the sense, the currents and the tension are readings. At this writing the engine on `main` still reads the holder of the sign by the reader's sign, which the owner's word of 2026-10-01 replaces by the plain read of the two-pair records in the round after it; the two experiments' worlds, which source no W , are unchanged by it; the paper describes those worlds as the implementation and cites no run of them. The rounds that follow, named by the decisions: the Node that knows no family (the width as the run's declaration, the Node stepping a flat list of lines of dimension one, the row's declarations in the file alone), the meeting round (the parts per family, the pair family with two, the instrument's joint share from the pair amplitude over the parts for every pair of ports, one draw per pair ordered in time, the joint-share Bell reader; its first look Bell on the same board against the blind $S = 2.83$), and the rotation round after Bell's look [2, the engine; the NodeState; how to add a family][3, the generic Node is closed for building; the engine works only with families of dimension one].

The files. *The universe file* declares the integers (Γ , T , the width) and the families, each its name, its pair and, for a held family, what it holds, its divisor and, on the massless row, its rest; the rule derives the rest. *The world file* declares the board's shape and its faces (periodic, open or closed; the inner faces with their gaps; the receding faces with their largest size and their layers per growth), the intervals, the bodies by their family, their Nodes with their counts, the messages by

their family, axis, wave number, amplitude, envelope, phase and transverse wave number, and the detectors by their regions or the body they name; a missing key is refused by name, and no default is written. *The mode file* beside the world holds the generator's levels of every body and message. *The expectation file* holds the blind row, every number the file's: the regions, the window, the blind counts and the draw's seed for a screen; the ports, the settings, the calibrated phases and the blind E and S for Bell. The two experiments' worlds are written by a builder from one design file each, every number of the worlds in it [2, the loader and the files; how to run a world].

A run. The generator lays the bodies and the messages, a record and no count; the gate at the start checks each body's declared count against its record's share, a reading and no lay; the run is headless, one output file per world with the verdict (LAWFUL, or REFUSED with the reason, the guard's and the gate's at load among them), the intervals run, the end where a receding face reached its largest size, one `click` line per family and region per interval with the net current through the region's front boundary (the detector's report, the measurement), a `field` line per region with a family's density over it (a GameBoard reading), and the books (the share's total, its drift against the one the world started with, the least Link pace of the final state, diagnostics). The readers: the screen's reader takes a world's `click` lines over the window against the expectation file and gives per region the share, N the total over W_c , the laid count less N as the part that left elsewhere, the clicks credited by the largest remainders and one draw of N clicks by the shares at the declared seed, DETECTOR; Bell's reader takes the runs' `click` lines per port, calibrates the ports and the settings' phases, credits each side by the rule the file names (the sign, or the contrast) and gives E among the coincidences, S , the efficiencies and the readings beside them in exact fractions; the two speeds' reader takes the `field` lines at the far region. A look runs only after the back-in-time gate says MATCH on its own world over its intervals [2, the output; how to run a world].

How to use the simulator, in a few lines. Write a world file: the board's shape and faces, the universe file it names, the bodies by their family, Nodes and counts, the messages by their family, axis, wave number, amplitude and envelope, and the detectors by their regions; every number of physics is in the files and none in the engine. Lay it: `PYTHONPATH=src python tools/pixel_mode.py --input world.json` writes the mode file with every body's fixed point and every message's levels. Run it headless: `PYTHONPATH=src python tools/run_inputs.py --out runs/first world.json` writes one output file with the verdict (LAWFUL, or REFUSED by name), the `click` lines (the measurement) and the `field` lines (a diagnostic). Check it: `PYTHONPATH=src python tools/back_in_time.py --intervals 40 world.json` runs the world forward and back and says MATCH or MISS by name; no look runs before MATCH. Read it: `tools/click_counts.py` credits a screen's clicks by the shares against the blind file, and `tools/look/record.py` with `tools/look/page.py` draws the board as a diagnostic. Where each chapter of this paper lives in it: the line in `core/rule3.py` and the Ports in `core/ports.py` (Section 2); the paces and the guard in `features/signed_read`; the readings of the form, the Wronskian, the currents and the tension in `node.py` and the share in `share.py` (Section 3); the one write in `features/hold` and `features/write` and the start in `features/start` (Sections 2.8 and 4); the detectors' reports in `reports.py` and the receding face in `growth.py` (Section 3.3); the interval and its inverse in `game_board.py` (Section 2.9); the families from the rule in `loader/derived.py` (Section 4); the generator in `tools/pixel_mode.py` (Section 5.5); and the two experiments' worlds under `examples/events/` (Section 7) [2, how to run a world].

The gates. Every pull request runs lint, formatting, strict types and the tests of the changed files and their consumers; the gates hold the rule's arithmetic to one function, every shift across a Link to the Ports, every integer literal of the engine and its tools to the law's own, no root anywhere,

every module to integers alone, the Node to the generic Node’s state and acts, the acts against Rule3 by hand, the interval on a closed cube (every holder at an isotropic rest, the 48 symmetries in every part, the gate MATCH), the share’s identity against the currents, the tension as Rule3’s own conservation of the current, light born by the write, the receding face on a chain of light alone, the rest of every pair as its line’s own fixed point, the generator’s bodies and its message lay against the real numbers, the inner face’s gap and reflection, the documents’ headings and the repository’s language. A behaviour change needs a dedicated test; a physics change starts from the law’s line [2, the gates].

To test a hypothesis. Write its line in the law’s document first, under its own name, with the three verdicts; the advisor computes the blind number from the engine’s own lines; the worker builds it as one pure function of arrays that calls Rule3 alone, with its test, on one short branch; the world and its blind expectation are written before the run; the run’s reading is compared with that number, a difference being a defect of the engine and never a finding, and a number outside its band with no defect found naming the missing law as a hypothesis under its own name. No expectation is rewritten after a look, and no body’s number is tuned to a result [21, the blind number first].

Statements and Declarations

Funding. None.

Competing interests. None.

Ethics approval, consent to participate and consent for publication. Not applicable; no human or animal participants.

Data and code availability. The law’s document, the engine that computes the rule and the figures’ scripts are archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [20], under the MIT licence; the paper cites no run; the worlds of its two experiments are in the archive as files, and their runs enter the paper when they have worked and been confirmed.

Author contributions. The author laid down every postulate and decision of the model, directed the work and wrote the paper.

The use of a large language model. The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under his direction and instructions, for the software, the computations, the mathematical derivations and their checks, the physics of the law’s document and the drafting of the text; the model is not an author; the author reviewed and edited every statement and takes full responsibility for the content. No figure was made by a generative tool; every figure is drawn by a script from the definitions or from the documents’ rows.

References

- [1] The law of the GameBoard, `docs/ALGEBRA.md` of the archive [20], cited by the name of its section: the GameBoard and the families (the objects, a family’s declaration, the interval, readings and measurements); Rule3 (the line, the direction, the four acts, the paces, the bound, the self-source, the conserved form); the click (the booking, the count is the record’s share, the click is the detector’s report, no click names a Node, the algebra of the exclusivity); the primitives; the stable body (what a body is, the well, the functional and the dilation, the surplus leaves, the generator, the velocity); what the law says of nature (the reading, with no waves on the event board, what produces a click is not known and nature’s numbers

enter the board only as clicks; the hypotheses under their own names; the postulates; the constants; the rows against nature); what is open.

- [2] The engine, `docs/ENGINE.md` of the archive [20]: the words, the NodeState, the interval, the loader and the files, the output, how to run a world, how to add a family, the gates.
- [3] The decisions, `docs/HIGHLIGHTS.md` of the archive [20]: the model owner's decisions in force, one line each, cited by the line's name.
- [4] J. S. Bell, On the Einstein Podolsky Rosen paradox, *Physics Physique Fizika* 1, 195 (1964), <https://doi.org/10.1103/PhysicsPhysiqueFizika.1.195>.
- [5] M. Born, Zur Quantenmechanik der Stoßvorgänge, *Zeitschrift für Physik* 37, 863 (1926), <https://doi.org/10.1007/BF01397477>.
- [6] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, Proposed experiment to test local hidden-variable theories, *Physical Review Letters* 23, 880 (1969), <https://doi.org/10.1103/PhysRevLett.23.880>.
- [7] J. G. Cramer, The transactional interpretation of quantum mechanics, *Reviews of Modern Physics* 58, 647 (1986), <https://doi.org/10.1103/RevModPhys.58.647>.
- [8] Y. Aharonov, P. G. Bergmann and J. L. Lebowitz, Time symmetry in the quantum process of measurement, *Physical Review* 134, B1410 (1964), <https://doi.org/10.1103/PhysRev.134.B1410>.
- [9] B. Hensen et al., Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres, *Nature* 526, 682 (2015), <https://doi.org/10.1038/nature15759>.
- [10] H. Yukawa, On the interaction of elementary particles. I, *Proceedings of the Physico-Mathematical Society of Japan* 17, 48 (1935), https://doi.org/10.11429/ppmsj1919.17.0_48.
- [11] B. P. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, Fermi Gamma-ray Burst Monitor and INTEGRAL), Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A, *The Astrophysical Journal Letters* 848, L13 (2017), <https://doi.org/10.3847/2041-8213/aa920c>.
- [12] G. H. Derrick, Comments on nonlinear wave equations as models for elementary particles, *Journal of Mathematical Physics* 5, 1252 (1964), <https://doi.org/10.1063/1.1704233>.
- [13] S. I. Pohozaev, Eigenfunctions of the equation $\Delta u + \lambda f(u) = 0$, *Soviet Mathematics Doklady* 6, 1408 (1965).
- [14] S. I. Pekar, *Untersuchungen über die Elektronentheorie der Kristalle* (Akademie-Verlag, Berlin, 1954).
- [15] C. M. Will, The confrontation between general relativity and experiment, *Living Reviews in Relativity* 17, 4 (2014), <https://doi.org/10.12942/lrr-2014-4>.
- [16] N. G. Vakhitov and A. A. Kolokolov, Stationary solutions of the wave equation in a medium with nonlinearity saturation, *Radiophysics and Quantum Electronics* 16, 783 (1973), <https://doi.org/10.1007/BF01031343>.
- [17] I. I. Shapiro, Fourth test of general relativity, *Physical Review Letters* 13, 789 (1964), <https://doi.org/10.1103/PhysRevLett.13.789>.
- [18] M. Giustina et al., Significant-loophole-free test of Bell's theorem with entangled photons, *Physical Review Letters* 115, 250401 (2015), <https://doi.org/10.1103/PhysRevLett.115.250401>.
- [19] L. K. Shalm et al., Strong loophole-free test of local realism, *Physical Review Letters* 115, 250402 (2015), <https://doi.org/10.1103/PhysRevLett.115.250402>.
- [20] A. Gonen, Universe24: a discrete simulator of local physical laws (Reality Theory), Zenodo (2026), <https://doi.org/10.5281/zenodo.22738746> (the concept DOI, resolving to the latest version).
- [21] The shared way of working, `skills/workflow.md` of the archive [20]: the three tests, the six verbs, the blind number first, the short procedure.